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Robustness of predictions in non-random samples

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The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam

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Executive summary

This thesis investigates the first steps in the robustness of predictions in non-random samples, specifically evaluating the predictive accuracy of the Heckman correction model under various conditions of sample selection bias. Sample selection bias is introduced through a practical example from e-commerce, highlighting the necessity of models that can correct for this bias in marketing related topics. The thesis provides a comprehensive review of the Heckman correction model, discussing its parametric assumptions, estimation methods (Maximum Likelihood and Two-Step), and the challenges it faces, such as sensitivity to distribution assumptions and collinearity among regressors.

The key focus of the thesis is on evaluating the predictive accuracy of the models, using metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), rather than just their explanatory power. Simulations are designed and implemented to compare the predictive accuracy of the Heckman models and a linear regression model (SSLR) under different conditions, including varying sample sizes, degrees of censoring, and correlations of error terms. The simulation results indicate that Heckman models generally have stronger predictive power compared to SSLR, particularly in scenarios with moderate to high selection bias. However, there is a threshold beyond which SSLR can outperform Heckman models. The real-world data on ambulatory expenditures further validates the findings, showing that Heckman models demonstrate superior predictive accuracy over SSLR.

The conclusion drawn from this research is that Heckman correction models offer improved predictive accuracy in the presence of sample selection bias. However, their performance is influenced by factors such as the degree of censoring, sample size and the correlation between error terms. This suggests that while the Heckman model is effective, there is potential for exploring alternative models and methods to further enhance predictive accuracy. Introducing more predictors reduced the bias committed by Heckman models and became closer to SSLR model. Recommendations for future research include testing predictions with a bigger number of explanatory variables and other scenarios such as other distribution assumptions of the error terms are considered, plus testing predictions in a marketing related domain.

1. Introduction

Imagine an ecommerce clothing brand wants to estimate sales based on customer behaviour interaction with the website. To do so, behavioural information from users is considered. Several explanatory variables can be used for prediction but to keep it simple let's assume dwell time is the only one used; and a positive influence on sales is found where the more time spent on the website, the higher the sales.

One has the problem that for some users no sale was registered. They are not considered in the model. This may lead to misleading conclusions because if only data of users who made a purchase is analysed, a significant portion of the customer behaviour data (those who didn't make a purchase) is ignored. This can lead to biased estimates because the sample of users who made a purchase is not random. It is a subset of all users who visited the website, and their behaviour (dwell time) might be systematically different from those who didn't make a purchase. Ignoring users who didn't make a purchase means that the factors that influence the decision to purchase are not being fully accounted. This can lead to overestimating the impact of dwell time on sales since those who already decided to buy are the only ones observed.

This is an example of what is called Sample Selection bias, a type of bias that occurs when the sample used in a study or analysis is not randomly selected from the population from which it was drawn. This bias can lead to incorrect conclusions because the results may not generalize to the broader population. It happens when certain groups are overrepresented or underrepresented in the sample, often due to a rule that selects the sample. In the previous example for instance, users who decided to buy might be loyal customers who tend to buy more while the others might be casual ones, or count on a lower income for example, differing from the others.

Selection bias can be present in different Marketing scenarios. The attribution model for instance can be biased as users who decided to install an ad blocker are not considered in the model (**Batista, 2017**). In this case the rule selecting the sample is based on what is called self-selection, that is, when individuals decide by themselves to participate or not. Another example where selection bias with self-selection is present is web surveys or reviews where users decide by themselves to answer the survey or not (**Bethlehem, 2010**).

Users with better Internet connection or with a very good experience with the brand might feel more encouraged to answer while the rest not, causing the sample to be non-random.

A method proposed to deal with selection bias was the Heckman correction; a statistical model that corrects for selection bias in observational data by simultaneously estimating two equations: one for the selection process (the probability of being included in the sample) and another for the outcome of interest. The classical Heckman model involves a parametric approach for estimation under the assumption of bivariate normality between the errors of those two equations (Heckman, 1974 and 1979). This led to the creation of more robust parametric and non-parametric methods in this matter in addition to other critics (Puhani, 2000). See Marchenko & Genton (2012) and Zhelonkin, Genton & Ronchetti (2016) for some parametric examples. Furthermore, see Vella (1998) and Zhelonkin & Ronchetti (2021) for reviews.

According to Shmueli (2010), there is a distinction between explanatory and predictive modelling because explanatory modelling seeks to understand underlying mechanisms and causal relationships, while predictive modelling aims to accurately forecast future outcomes, and these objectives require different approaches and validation methods due to the gap between theoretical constructs and measurable data. The focus of this thesis is on predictions. Existing literature works improving the classical Heckman models, but it is focused on parameter estimation and hypothesis testing, more focused on explanatory modelling and not considering the predictive accuracy of the models. The aim of this thesis is to test, under several scenarios and circumstances, the predictive accuracy of the classical Heckman methods when predicting under sample selection bias, compared to one linear regression considering only selected individuals (**SSLR – *selected sample linear regression* –**) acting as model that is not correcting for sample selection bias. Results show that Heckman models count on a stronger predictive power than SSLR until a certain threshold where this latter, even if only considers selected individuals, becomes more accurate in its predictions.

The manuscript is structured as follows. Section 2 offers a review of the classical Heckman models. Section 3 summarizes the measures for evaluations predictions. Sections 4 and 5 summarizes the results obtained from empirical application on predictions with

simulated data build under different scenarios. Section 6 test for predictions with real data, the Ambulatory expenditures data used in the book by **Cameron and Trivedi (2009, pp.541 -550)** is considered as a business application. Section 7 concludes.

2. Classical Heckman model overview

The standard Heckman model (or Type 2 Tobit model) consists of two equations: an outcome and a selection equation. There is an outcome equation (1.a) that represents the main regression model for predictions.

$$y_{1i}^* = x_{1i}'\beta_{1i} + u_{1i}, \quad (1.a)$$

where y_{1i}^* is the outcome variable, x_{1i}' is a vector of explanatory variables, β_{1i} is a vector of coefficients and u_{1i} is the error term. However, the existence of some selection bias/rule (1.b) makes that only observations following this selection mechanism (1.c) take part in the outcome equation while not considering the rest, setting a non-random sample not representative of the population of interest.

$$y_{2i}^* = x_{2i}'\beta_{2i} + u_{2i}, \quad (1.b)$$

$$y_{1i} = \begin{cases} 0 & \text{if } y_{2i}^* = 0, \\ y_{1i}^* & \text{otherwise.} \end{cases} \quad (1.c)$$

y_{2i}^* is a binary latent variable representing the propensity to be selected, x_{2i}' is a vector of explanatory variables, β_{2i} is a vector of coefficients and u_{2i} is the error term. Unobserved characteristics from outcome and selection equation represented by u_{ji} come from a bivariate normal distribution (2), the parameter ruling the selection bias is $\rho \in (-1, 1)$, generating bias when $\rho \neq 0$. And vectors β_{ji} are unknown parameters.

$$\begin{pmatrix} u_{1i} \\ u_{2i} \end{pmatrix} \sim BN \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} \right) \quad (2)$$

To account for selection bias within this scenario, two main estimation methods are formulated by Heckman. Based on the errors' distribution assumption, the most efficient

method is **Maximum likelihood estimation (ML)**. By jointly estimating the selection equation and the outcome equation, it accounts for the probability of selection into the sample when estimating the parameters of the outcome equation. This is done by maximizing the likelihood function of model (1) shown below (3) (**Amemiya, 1985, p. 386**):

$$L = \prod_{y_1=0} 1 - \Phi\left(\frac{x'_{2i}\beta_2}{\sigma_2}\right) \prod_{y_1>0} \Phi\left\{\left(x'_{2i}\beta_2 + \frac{\sigma_{12}}{\sigma_1^2}(y_1 - x'_{1i}\beta_1)\right)\sqrt{\sigma_2^2 - \frac{\sigma_{12}^2}{\sigma_1^2}}\right\} \frac{1}{\sigma_1} \phi\left(\frac{(y_1 - x'_{1i}\beta_1)}{\sigma_1}\right). \quad (3)$$

Due to power/time consuming constraints in the past taken by ML method in addition to its inefficiency when the distribution assumption of errors does not hold, a second estimation method proposed is the **Two step method (2step)**, a two-step process where first the selection equation is estimated using a probit model, modelling the probability of selection into the sample; and then adding as an additional regressor to the outcome equation (6) the Inverse Mills Ratio (IMR), a function of the estimated coefficients from the selection equation that represents the expected value of being selected into the sample given the observed characteristics that corrects for selectivity. The IMR (5) is obtained by taking the conditional expectation of the selection rule, the right part of (4) which is the conditional expectation of y_{1i}^* . The outcome equation (6) is then estimated through ordinary least squares method (OLS).

$$E(y_{1i}^* | x_{1i}, y_{2i}^* > 0) = x'_{1i}\beta_1 + E(u_{1i} | u_{2i} > -x'_{2i}\beta_2), \quad (4)$$

$$\lambda\left(\frac{x'_{2i}\beta_2}{\sigma_2}\right) = \frac{\phi\left(-\left(\frac{x'_{2i}\beta_2}{\sigma_2}\right)\right)}{1 - \Phi\left(-\left(\frac{x'_{2i}\beta_2}{\sigma_2}\right)\right)}, \quad (5)$$

$$y_{1i}^* = x'_{1i}\beta_1 + \lambda\left(\frac{x'_{2i}\beta_2}{\sigma_2}\right)\rho\sigma_2 + \varepsilon_1 = x'_{1i}\beta_1 + IMR \beta_\lambda + \varepsilon_1. \quad (6)$$

The main critics to Heckman's estimators (**Puhani, 2000**) are the fact that they become more sensitive as the distribution assumption of error terms disappears (specially with ML) and secondly, the sensitivity of the two methods to the collinearity among regressors in the primary and selection equations. The problem of collinearity arises because the identification

of the regression parameters relies on the nonlinearity of the inverse Mills ratio (IMR). The IMR can exhibit near-linearity over a wide range of its support. This near linearity reduces the variation provided by the IMR, leading to high collinearity with other explanatory variables in the outcome equation. This can be possible as it is normal to find similar variables -even identical- in the outcome and selection equation. Due to this, it is expected to have at least one variable in the selection equation that is not included in the outcome equation for the model to work well (Little and Rubin, 1987, p. 230).

Additionally, Puhani (2000) states as a result of several MonteCarlo simulations that in cases where high censoring and correlation of errors is present, that is, when there is high selection bias, Heckman estimators perform poorly compared to OLS.

The goal is to predict $\widehat{y}_1$ (7) representing the predicted value of the outcome variable y_1 under several scenarios and situations criticized in the literature to analyse the predictive accuracy of Heckman models against a regression only considering selected individuals (SSLR).

$$\widehat{y}_{1i} = x'_{1i}\widehat{\beta}_{1i} + \widehat{IMR}\widehat{\beta}_{\lambda}, \quad (7)$$

where $\widehat{y}_1$ is the predicted value of the dependent variable, x'_1 is a vector of explanatory variables, $\widehat{\beta}_1$ is a vector of estimated coefficients, IMR is the inverse mills ratio and $\widehat{\beta}_{\lambda}$ the estimated coefficient impact of the selection bias correction on the outcome.

3. Predictions evaluation

The aim when testing predictive power is generalization, that is, models trained on a specific dataset might perform well on that data but fail to generalize to new, unseen data **Shmueli (2010)**. That is why data is partitioned into a training set where the models are trained and a holdout data to test and compare predictive accuracy between models.

To comprehensively assess the performance of our predictive models, three evaluation metrics are implemented: Mean Squared Error (MSE), Root Mean Squared Error (RMSE)

and Mean Absolute Error (MAE). These metrics provide complementary perspectives on model accuracy and error distribution, enabling a robust comparison of model performance.

Mean Squared Error (MSE): MSE is the mean of the squared differences between predicted and actual values. It emphasizes larger errors, offering a clear measure of prediction accuracy by highlighting significant deviations.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_{1i} - \widehat{y}_{1i})^2 . \quad (8)^1$$

Root Mean Squared Error (RMSE): RMSE is the square root of the average squared differences between predicted and actual values. It penalizes larger errors more than MAE, making it sensitive to outliers and providing insights into the distribution of error magnitudes.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{1i} - \widehat{y}_{1i})^2} . \quad (9)^1$$

Both metrics show the average error made in the predictions, but both are needed. El RMSE takes the square root of MSE, converting the result back into the original units of the data so this offers a clearer more realistic vision of how much predictions deviate on average. This is helpful for instance when analysing predictions on Ambulatory data as results can be interpreted in the same units of the predicted variables, that is, in dollars. On the other hand, MSE is useful because it can be decomposed into the square bias and the variance of residuals (10) offering a vision of the bias-variance trade off. Bias refers to the error from overly simplistic models that fail to capture the true patterns in the data, resulting in underfitting. That is, the performance of the Heckman models with a biased model must be compared not only in terms of bias but also of variance.

$$MSE = \quad Bias^2 \quad + \quad Variance \quad + \quad Irreducible \text{ Error} , \quad (0.1)$$

$$MSE = (E[\widehat{y}_{1i}] - y_{1i})^2 + E[(\widehat{y}_{1i} - E[\widehat{y}_{1i}])^2] + \sigma^2 . \quad (10.2)$$

Mean Absolute Error (MAE): MAE measures the average magnitude of errors in predictions, calculated as the average of the absolute differences between predicted and actual values (11). It is easy to interpret as it is in the same units as the target variable and is less

¹ n represents the sample size. It represents the full sample, with observed and unobserved individuals unless predictions are made under sample selection bias. In that case it represents only selected individuals. y_{1i} is the actual outcome value for the i-th data point and $\widehat{y}_{1i}$ is the predicted outcome value for the i-th data point.

sensitive to outliers, treating all errors equally. RMSE (Root Mean Squared Error) also measures prediction accuracy but squares the errors before averaging them, giving more weight to larger errors. RMSE will always be equal to or greater than MAE due to the squaring of errors, and it reflects the standard deviation of prediction errors. The choice between MAE and RMSE depends on whether the focus is on penalizing larger errors more heavily (favoring RMSE) or treating all errors equally (favoring MAE).

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_{1i} - \widehat{y}_{1i}|. \quad (11)^1$$

As a final remark, to get a more reliable estimate of the model's performance, predictions are run multiple times. That is why throughout the manuscript, when talking about error values, it is said to be the average RMSE, average MSE or average MAE.

4. Simulation Design

The simulations considered are summarized in Table 4.1. Predictions are run under three different scenarios (detailed explanation in Table 4.2): one where there is no exclusion restriction (scenario 3), and two others with exclusion restriction, one where the predictor of the outcome equation is also included in the selection equation (scenario 2) and a second where is not (scenario 1).

Four methods are trained and tested: an ordinary least square linear regression representing predictions with only observed/selected individuals (**SSLR** – selected sample linear regression), Heckman Maximum Likelihood (**ML**), Heckman 2step (**2step**), and the true relationship model, coming from an OLS model considering both observed and unobserved individuals. This true relationship is named as ‘**True**’.

These models are trained and tested over the different scenarios under different sample sizes, degrees of censoring, and correlation of error terms and variables to analyse how prediction accuracy may vary. A total of 108 predictions for each model are done over a loop of 100 iterations to ensure stability and robustness of conclusions.

¹ n represents the sample size. It represents the full sample, with observed and unobserved individuals unless predictions are made under sample selection bias. In that case it represents only selected individuals. y_{1i} is the actual outcome value for the i -th data point and $\widehat{y}_{1i}$ is the predicted outcome value for the i -th data point.

Models analysed	Estimators used	Sample size	Repetitions	Distributions of u_2, u_1	Variables changed	Judgement criteria for predictions
Scenario 1	ML	2300 (2004 for training & 296 for testing)	100	Biv. normal	$Cor(u_2, u_1) = 0, -0.5, -0.85$	MSE (and decomposed MSE)
Scenario 2	2step					
Scenario 3	SSLR	800 (500 for training & 300 for testing)			$Cor(var_2, var_1) = 0, -0.5, -0.85, -0.95$	RMSE
	True				$Degree\ of\ censoring = 30\%, 80\%$	

TABLE 4.1: *Combinations for predictions.*

Scenario 1	Scenario 2	Scenario 3
$y_2 = var2'_i \beta_i + u_{2i}$ $y_1 = var1'_i \beta_i + u_{1i}$	$y_2 = var1'_i \beta_i + var2'_i \beta_i + u_{2i}$ $y_1 = var1'_i \beta_i + u_{1i}$	$y_2 = var1'_i \beta_i + u_{2i}$ $y_1 = var1'_i \beta_i + u_{1i}$
$y_2 = \begin{cases} 0 & \text{if } y_2 < 0 \\ 1 & \text{otherwise} \end{cases}$ $y_1 = \begin{cases} 0 & \text{if } y_2 = 0 \\ y_1 & \text{otherwise} \end{cases}$	$y_2 = \begin{cases} 0 & \text{if } y_2 < 0 \\ 1 & \text{otherwise} \end{cases}$ $y_1 = \begin{cases} 0 & \text{if } y_2 = 0 \\ y_1 & \text{otherwise} \end{cases}$	$y_2 = \begin{cases} 0 & \text{if } y_2 < 0 \\ 1 & \text{otherwise} \end{cases}$ $y_1 = \begin{cases} 0 & \text{if } y_2 = 0 \\ y_1 & \text{otherwise} \end{cases}$

TABLE 4.2: *Scenarios considered for predictions.*

In the code below², the creation of Scenario 1 is shown as an example. A sample size is set equal to 800. Error terms are generated from a multivariate normal distribution and are negatively correlated at with $\rho = 0.85$. Explanatory variables (v_1 and v_2 , which later will be named $var1$ and $var2$) are also generated from a multivariate normal distribution but then transform to uniform. The first step is done to have correlation between them if needed. In this case is set to -0.85.

Variables are stored in a data frame named ‘data’. Response variable (y_{1i}) ‘latent.truth’ is added using the explanatory variable ‘var1’ and the error term ‘errors.1’. The

² The lines of code shown are extracts from the entire code. See Appendix for detailed information about the code.

selection restriction is introduced using ‘var2’ and ‘errors.2’ and imposed in the response variable creating a ‘bias.outcome’ response variable where dependent variable from not selected individuals is set to 0. A summary of the first five observations resulting from this data set is shown in Table 4.3.

```
R> library(mvtnorm)
R> set.seed(22)

R> correlated_normals <- mvrnorm(800, c(0, 0), matrix(c(1, -0.85, -0.85, 1), 2, 2))
R> v1 <- pnorm(correlated_normals[,1]) # Transform to uniform
R> v2 <- pnorm(correlated_normals[,2]) # Transform to uniform
R> errors <- rmvnorm(800, c(0, 0), matrix(c(1, -0.85, -0.85, 1), 2, 2))

R> data <- data.frame(var1 = v1, var2 = v2, errors = errors)
R> data$selection <- data$var2 + data$errors.2 > 0
R> data$latent.truth <- data$var1 + data$errors.1
R> data$bias.outcome <- data$latent.truth * (data$selection > 0)
```

var1	var2	errors.1	errors.2	selection	latent.truth	bias.outcome
0.9997	0.2304	0.6508	0.5743	TRUE	1.5740	1.5740
0.9997	0.2304	0.6508	0.5743	TRUE	1.5740	1.5740
0.9964	0.0662	-1.1937	-0.8286	FALSE	0.1678	0.0000
0.9863	0.3132	-2.3069	1.6890	FALSE	2.6753	0.0000
0.9848	0.8935	2.5249	-2.9210	TRUE	-1.9361	-1.9361

TABLE 4.3: First 5 observations from ‘data’ data frame from one of simulations with scenario 1.

The data set ‘data’ is divided into a training set (**train**) and a testing/validating set (**test**). However, two more splits are done: a training set with selection (**train_ws**) coming from only selected observations in ‘train’ set, and a test set with selection (**test_ws**) coming from only selected observations in ‘test’ set.

```
R> library(caret)
```

```

R> for (i in 1:100) { #simulations are done over a loop of 100 iterations
R>   set.seed(i)
R>   train_indices <- createDataPartition(data$var1, p = 0.625, list = FALSE)
      #partition weight: 0.625 with small sample (n = 800) and 0.869 with big sample (n = 2300)
R>   train <- data[train_indices, ]
R>   test <- data[-train_indices, ]
R>   train_ws <- train[train$selection, ]
R>   test_ws <- test[test$selection, ]
R>   ...3

```

This is due to the following. Predicting with the SSLR model simulates the case when predicting with selection bias. That means it only considers observed individuals thus, the model can only be trained with a data set with selection (`train_ws`). In the case of Heckman models, they considered the explanatory information from unobserved individuals that will help to correct for that selection bias thus, a data set with selected and not selected individuals must be considered (`train`). And for the True model, the true relationship, it is logical to consider all individuals thus, it is trained using ‘train’ set.

True and SSLR models use `latent.truth` as the dependent variable but Heckman models use `bias.outcome` as the dependent variable for non-selected observations must be 0 or NA. Heckman models are trained with function ‘selection’ from the **sampleSelection** package by **Toomet and Henningsen (2008)**.

```

R> library(sampleSelection)
R> model.true <- lm(latent.truth ~ var1, data = train)
R> model.ols <- lm(latent.truth ~ var1, data = train_ws)
R> model.ml <- selection(selection ~ var2, bias.outcome ~ var1, method = "ml", data = train)
R> model.2step <- selection(selection ~ var2, bias.outcome ~ var1, method = "2step", data = train)

```

When it comes to making predictions in the presence of selection bias, that means that they can only be tested with selected individuals as these are the only ones that counts on information about the dependent variable y_{1i} . Thus, all models are tested in ‘test_ws’. Nevertheless, as this study is done with simulated data, the true picture is present, that is, information about y_{1i} for not selected individuals is available so, also running predictions in

³ The lines of code shown are extracts from the entire code. See Appendix for detailed information about the code. It is disposed as a loop to run train, predict and compute the metrics in one run.

‘test’ data set is possible, allowing to check the robustness of the models with the true picture. This is summarised in Figure 4.1.

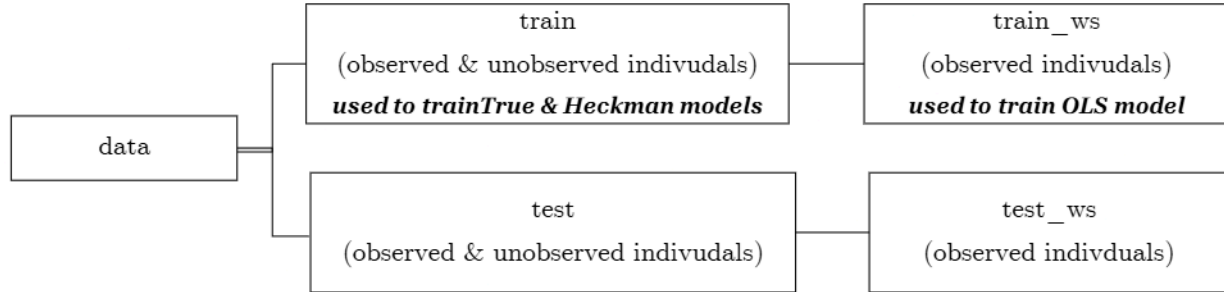


FIGURE 4.1: Data partition flow for training and validating sets

5. Simulation results

The results when predicting in a sample with selection (‘test_ws’) offer different insights. In scenario 1, the predictions made with the two models with Heckman estimators always show on average a larger error compared to the SSLR estimator. Furthermore, as selection bias increases, the difference in error between the Heckman and SSLR models increases. This is shown in Figure 5.2. The graphs show the root mean square error (RMSE) of the predictions of the three trained models, for four different levels of correlation between the variables var1 and var2. Each row of graphs corresponds to a sample size and a degree of censoring, and each column at different levels of selection bias. Not only the degree of selection bias but also a greater degree of censoring enhances this difference of error between models. Furthermore, it can be observed that when the sample is smaller, there is a tendency for the error made by the Heckman models to increase as the correlation between variables is greater, while in a larger sample this trend is not as notable.

On the other hand, when decomposing the mean square error (MSE), the bias-variance trade off shows how, on average, the variance of the residuals remains the same or similar between models and what determines the change in the error is the bias (square bias of residuals) so when making predictions it is a matter of bias and not variance. In this case

Heckman models have higher bias than SSLR. Some of the metrics from some predictions are summarized in Table 5.1⁴.

Especially the 2step method is the most biased. Only when there are high levels of selection bias and the sample is large, the ML method has on average a greater error in its predictions at times, the rest of the times it is the 2step method, which, at the same time, fluctuates more. This means that of the 100 iterations on which the predictions are made, when observing the distribution of errors (MSEs), the 2step method commits on average a greater error than ML but it also makes errors with wider values than those of the ML method, a greater variance in the errors made, so it is possible to find predictions with more extreme errors when predicting with 2step than with ML estimator. This situation is not present when the sample is large and the degree of censorship small, but rather appears when the degree of censorship increases and when the sample is smaller (see Figure 5.1 for an example).

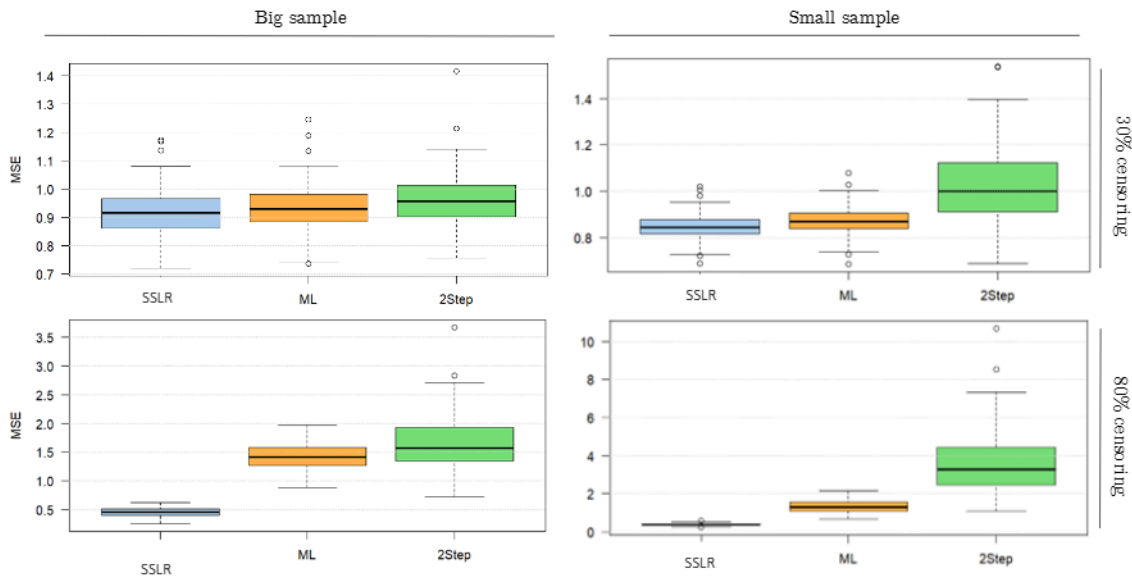


FIGURE 5.1: *Distribution of MSEs from 100 iterations for each model in scenario 1. Top-left [corr.errors = -0.5, corr.var = -0.95, $N = 2300$, censoring = 30%], top-right [corr.errors = -0.5, corr.var = -0.85, $N = 800$, censoring = 30%], bottom-left [corr.errors = -0.85, corr.var = -0.95, $N = 2300$, censoring = 80%], bottom-right [corr.errors = -0.85, corr.var = -0.85, $N = 800$, censoring = 80%].*

⁴ The metrics correspond to the same predictions shown in Figure 5.2. For entire visualization of all prediction metrics see Appendix

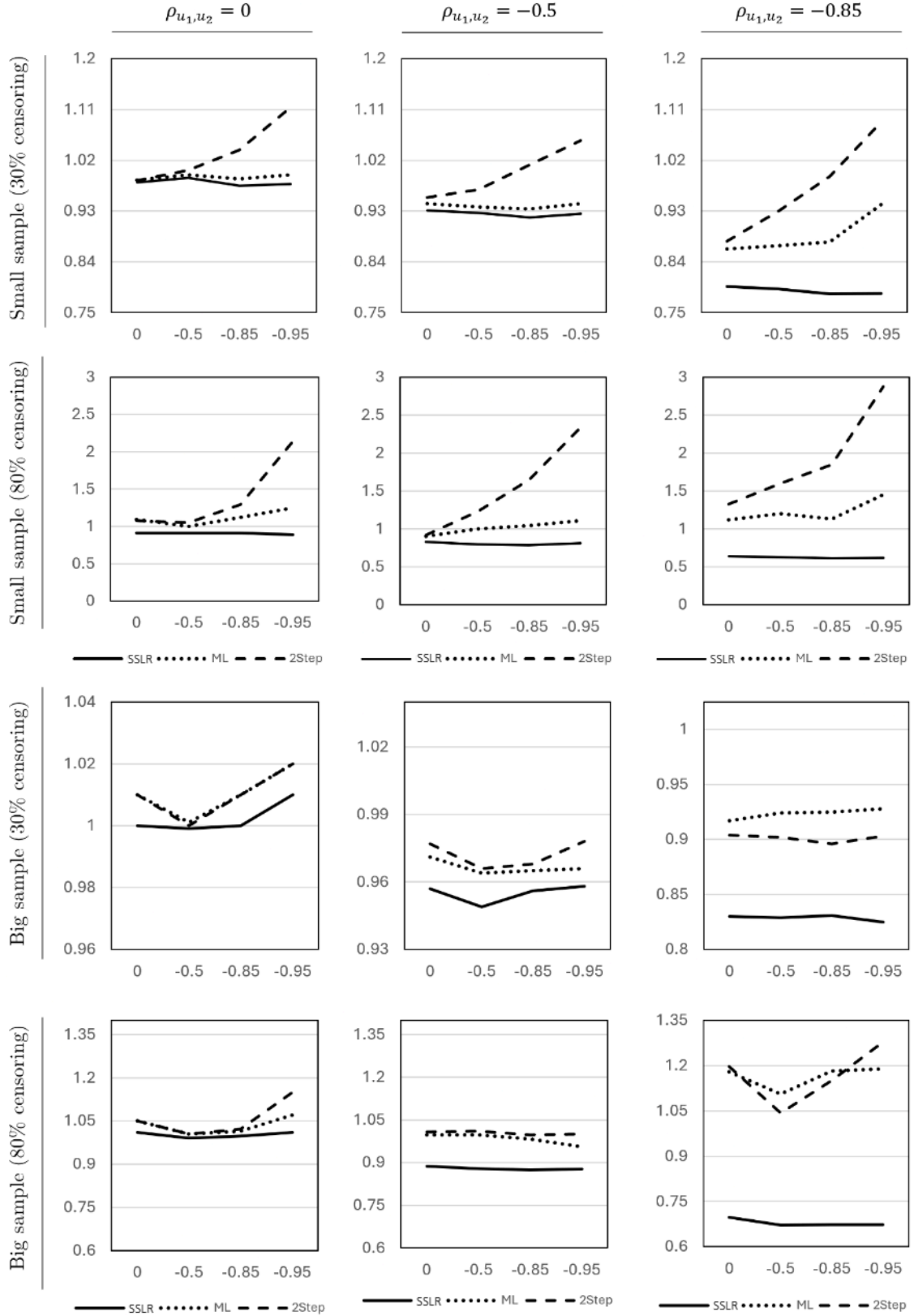


FIGURE 5.2: average RMSE resulting from 100 prediction iterations for the three different models in Scenario 1. y-axes represent different values corresponding to the average RMSE. x-axes represent different values of correlation between var1 and var2.

corr.errors = -0.5, corr.var = -0.95, N = 2300, censoring = 30%								
Model	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE Intercept	Var1	SE_Var1
SSLR	0.958	0.92	0.92	0.0048	-0.141	0.049	0.804	0.088
ML	0.966	0.935	0.921	0.0188	0.148	0.058	0.991	0.108
2Step	0.978	0.958	0.922	0.0398	0.148	0.058	0.989	0.108
corr.errors = -0.85, corr.var = -0.95, N = 2300, censoring = 80%								
SSLR	0.673	0.456	0.454	0.0082	-0.908	0.057	0.421	0.119
ML	1.189	1.421	0.483	0.9453	-1.375	0.07	1.141	0.113
2Step	1.278	1.664	0.488	1.1828	-1.376	0.07	1.144	0.114
corr.errors = -0.5, corr.var = -0.85, N = 800, censoring = 30%								
Model	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE Intercept	Var1	SE_Var1
SSLR	0.919	0.846	0.844	0.006	-0.13	0.093	0.809	0.167
ML	0.934	0.874	0.845	0.0321	0.147	0.116	1.087	0.22
2Step	1.012	1.03	0.855	0.179	0.157	0.117	1.066	0.221
corr.errors = -0.85, corr.var = -0.85, N = 800, censoring = 80%								
SSLR	0.615	0.381	0.377	0.0102	-0.906	0.111	0.532	0.233
ML	1.131	1.3	0.396	0.9111	-1.503	0.141	1.232	0.225
2Step	1.846	3.568	0.433	3.1423	-1.456	0.145	1.154	0.234

TABLE 5.1: summary of average metrics extracts from 4 different prediction iteration cases in scenario 1. From top to bottom they are the same predictions from Figure 3.3.

The difference between scenario 1 and scenario 2 is the presence of the variable 'var1' in both the selection and outcome equations in scenario 2, while in scenario 1 each equation has a different variable. The predictions in scenario 2 also show that on average the error made by the Heckman models is always higher than that made by the SSLR model (Figure 5.3 in next page).

The pattern that follows is similar, as the selection bias and the degree of censoring increase, the errors made between the Heckman and SSLR models are separated on average. The trend with a small sample where the RMSE of the Heckman models increases compared to that of the SSLR as the correlation between variables increases (first two rows of Figure 5.3) is significantly greater than in scenario 1 and this trend is already seen more present in situations with a larger sample.

On this occasion, the ML model tends to make more errors or equal that of the 2step when the sample increases (bottom right graphs in Figure 5.3). In small samples where the 2step has a higher RMSE, a situation similar to that of scenario 1 occurs when looking at the distribution of 100 iterations, when there is greater censoring and more correlation between the variables, the 2step model commits on average a greater error than ML but with greater variability, being able to predict with wider error values than those of ML. However,

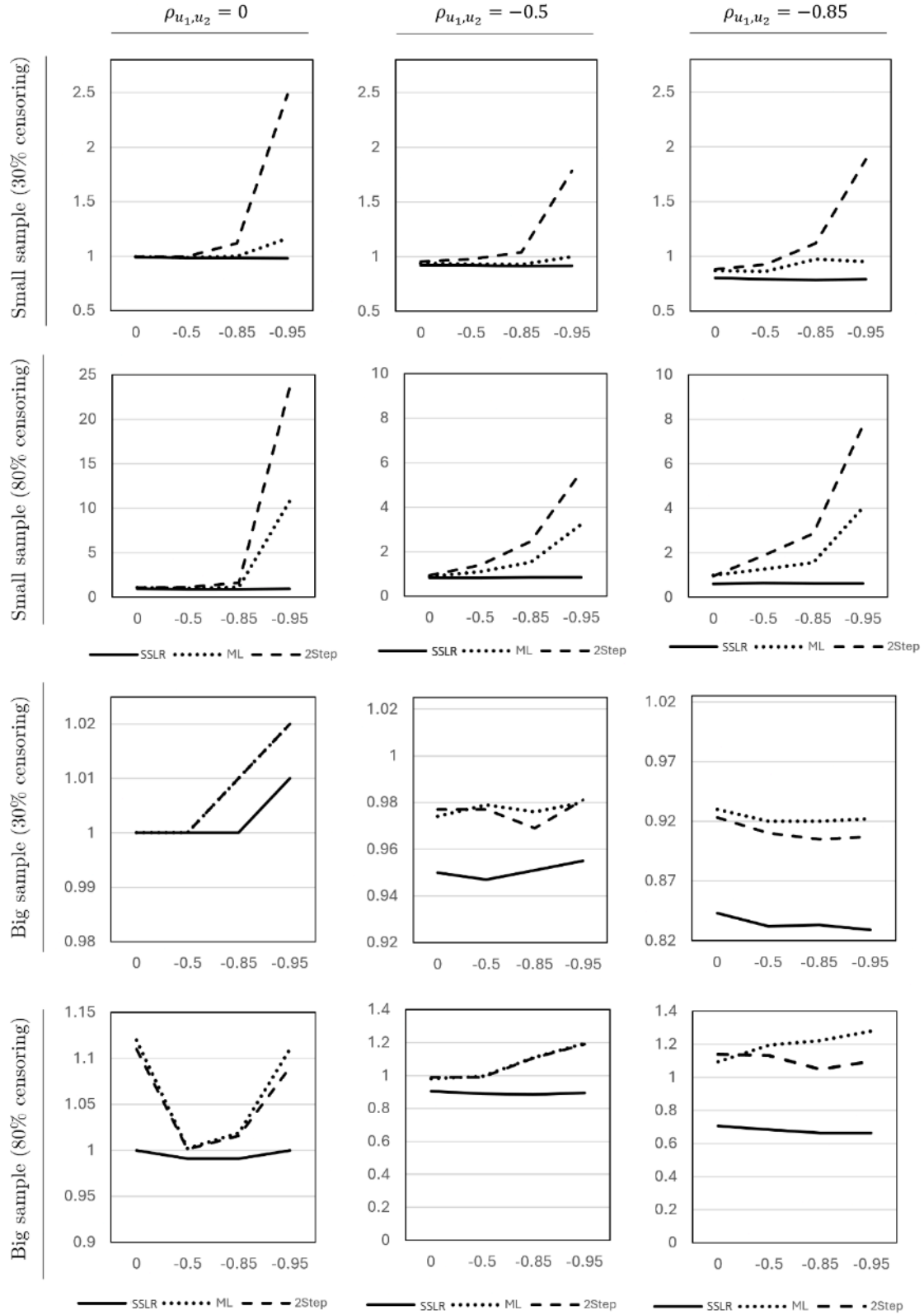


FIGURE 5.3: average RMSE resulting from 100 prediction iterations for the three different models in Scenario 2. y-axes represent different values corresponding to the average RMSE. x-axes represent different values of correlation between var1 and var2

this time more extreme values are observed, predictions that make an unusual error, becoming more visible outliers (see Figure 5.4).

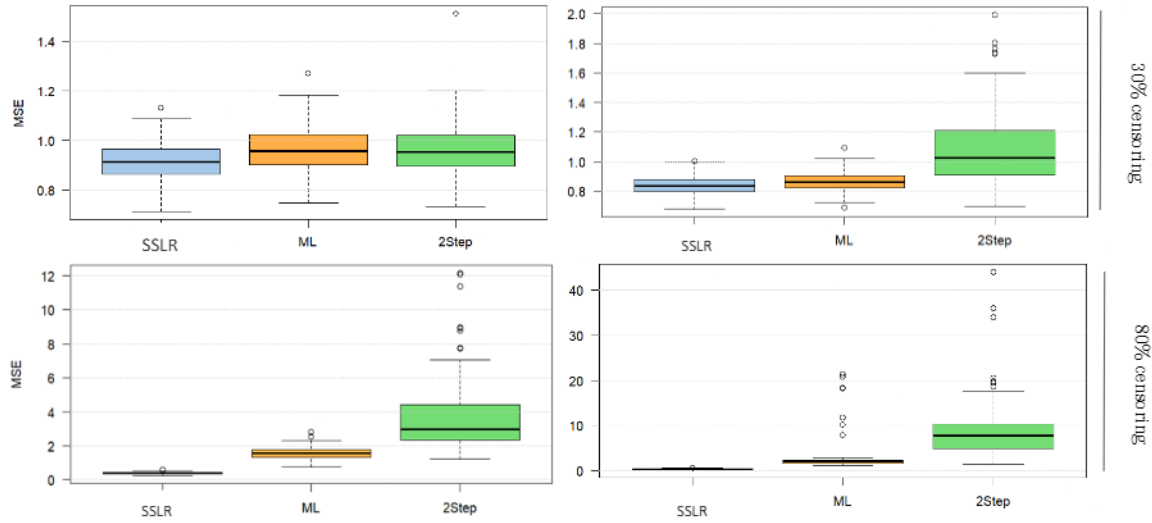


FIGURE 5.4: *Distribution of MSEs from 100 iterations for each model in scenario 2. Top-left [corr.errors=-0.5,corr.var=-0.95,N=2300, censoring = 30%], top-right [corr.errors = -0.5, corr.var = -0.85, N = 800, censoring = 30%], bottom-left [corr.errors = -0.85, corr.var = -0.5, N = 800, censoring = 80%], bottom-right [corr.errors = -0.85, corr.var = -0.85, N = 800, censoring = 80%].*

Finally, scenario 3 does not have exclusion restriction unlike the previous scenarios. In this case, the Heckman models also show a larger error on average than those of the SSLR model, although there is no longer a pattern like before (see Figure 5.5)⁵. This time, the 2step model is the one that always commits the most error on average and when the degree of censoring is high, this error committed is on average significantly greater than in the previous scenarios. However, when censoring is high, although the error is mainly committed by a high bias compared to that of the SSLR model, the variance also increases, so the errors committed are on average more dispersed than in those of the SSLR model (see Table 5.2 for an example).

Furthermore, of the 100 iterations on which the predictions are made, the errors made by the 2step models take more dispersed values than those of the ML to a greater extent than in the previous scenarios and when the sample decreases, errors with more extreme values outside the normal ones are observed (see Figure 5.6).

⁵ Unlike Figures 5.2 and 5.3, which is a two-dimensional representation, this time it is only one dimension since in scenario 3 only one variable is use for both selection an outcome equation.

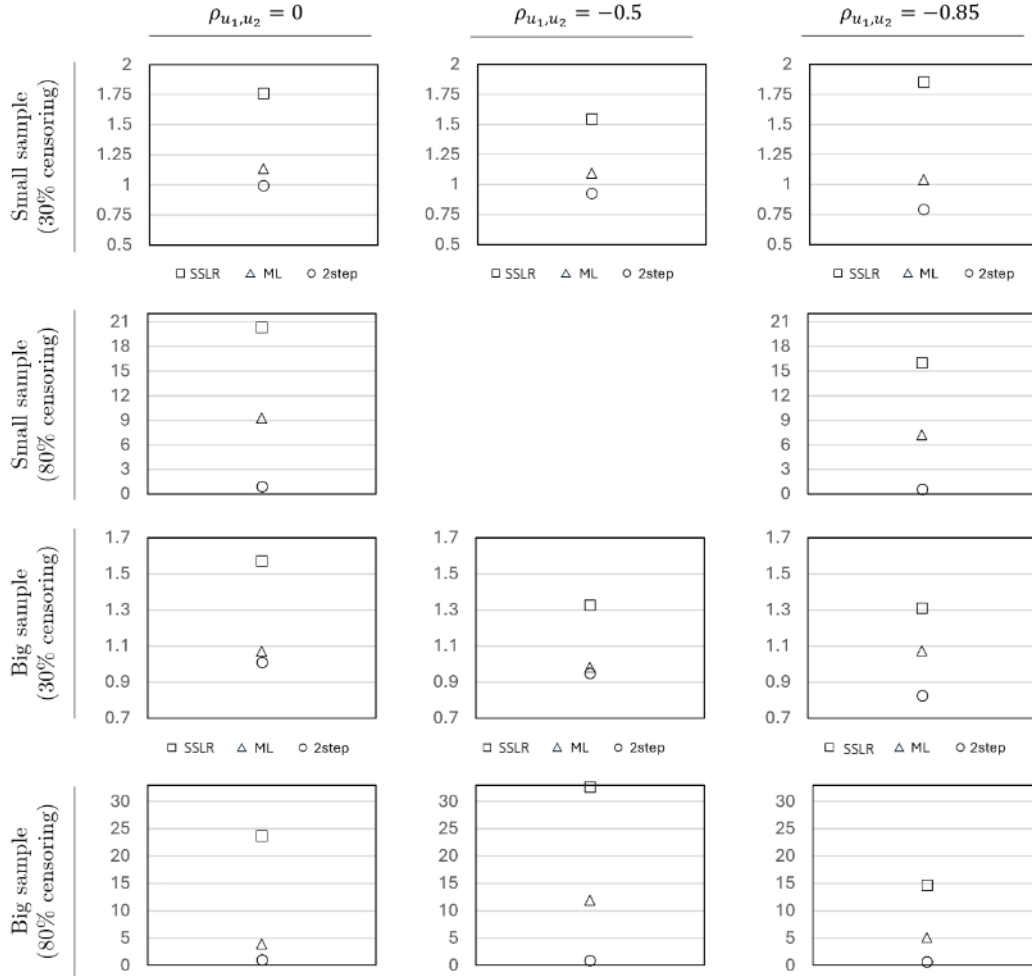


FIGURE 5.5: average RMSE resulting from 100 prediction iterations for the three different models in Scenario 3. y-axes represent different values corresponding to the average RMSE.

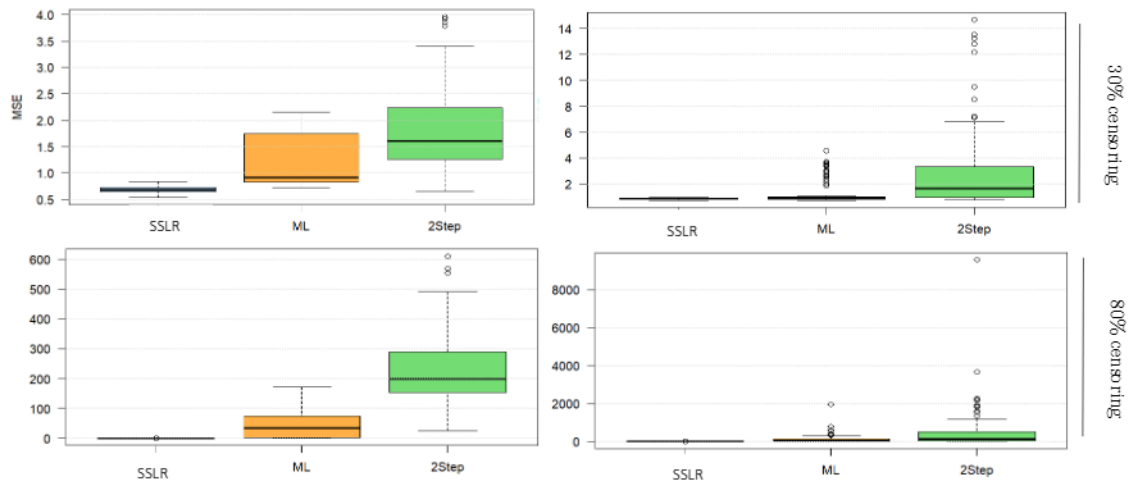


FIGURE 5.6: Distribution of MSEs from 100 iterations for each model in scenario 3. Top-left [corr.errors = -0.85, $N = 2300$, censoring = 30%], top-right [corr.errors = -0.5, $N = 800$, censoring = 30%], bottom-left [corr.errors = -0.95, $N = 800$, censoring = 80%], bottom-right [corr.errors = -0.95, $N = 800$, censoring = 80%].

corr.errors = -0.85, N = 2300, censoring = 30%								
Model	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.826	0.684	0.684	0.00322	-0.604	0.047	1.393	0.076
ML	1.071	1.191	0.735	0.46003	0.159	Inf	0.948	Inf
2Step	1.309	1.795	0.763	1.03559	0.164	0.058	0.933	0.107
corr.errors = -0.5, N = 800, censoring = 30%								
SSLR	0.926	0.858	0.856	0.00638	-0.356	0.101	1.25	0.165
ML	1.093	1.293	0.973	0.32478	0.171	Inf	1.01	Inf
2Step	1.545	2.923	1.045	1.88189	0.166	0.115	1.02	0.214
corr.errors = -0.95, N = 800, censoring = 80%								
Model	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.673	0.456	0.454	0.0097	-1.534	0.08	1.593	0.121
ML	5.02	39.16	4.866	34.4	-1.304	Inf	0.923	Inf
2Step	14.683	231.2	5.382	226	-1.308	0.07	0.942	0.114
corr.errors = -0.95, N = 800, censoring = 80%								
SSLR	0.587	0.347	0.343	0.0105	-1.642	0.142	1.867	0.214
ML	7.19	109.2	12.793	96.6673	-1.372	Inf	0.848	Inf
2Step	15.975	505.0	7.575	497.5858	-1.361	0.142	0.855	0.229

TABLE 5.2: summary of average metrics extracts from 4 different prediction iteration cases in scenario 3. From top to bottom they are the same predictions from Figure 3.7.

In summary, when predicting in a sample with selected individuals using the classic Heckman correction models, they will always predict with a greater error compared to if predicted with an SSLR model. This error is produced by a high bias and when it is a scenario without exclusion restriction, it is not only produced by high bias but also high variance as the degree of censoring increases.

Between the 2step and the ML estimator, predicting with a 2step model produces a greater mean error in its predictions in most cases but at the same time it can offer predictions with more varied error levels when the sample decreases and the censoring degree and the correlation between variables increases, so it is possible to find predictions with a greater or lesser error than when predicting with ML estimator.

There seems to be a relationship where including variables from the outcome equation in the selection equation to the point that there is no exclusion restriction causes a greater average error when predicting with the Heckman models, especially with the 2step estimator and especially in high cases of selection bias, degree of censoring and correlation between variables.

That predicting with Heckman models produces on average more error than predicting with the SSLR model may be due to two reasons. Firstly, the SSLR model makes less error on average since it has been trained with selected individuals, while the Heckman models include selection bias correction, offering a model closer to the true relationship that includes non-selected individuals which are not represented when predicting in 'test_ws'. Secondly, Heckman models can produce more error on average because they are not robust enough and the SSLR model, even when trained with selected individuals, is closer to the true relationship.

This can be verified by making predictions with simulations with a representative sample since there are values for the dependent variable of the non-selected individuals (something that does not happen in reality). Figure 5.7 shows the same metrics as in the previous figures but this time they represent the mean MSE of predictions made with a sample with selected and non-selected individuals ('test'). Only those cases are shown where the Heckman models perform worse than the SSLR model, that is, when the mean MSE is lower for it compared to the Heckman models.

It can be seen that when there is no selection bias, the SSLR model performs better for all scenarios. For scenarios 1 and 2 when the sample is small, as the bias selection increases, the less robust the SSLR model is, and the Heckman models perform better compared to it. However, high levels of correlation between variables can still make Heckman models less robust. This is the case of the 2step model in scenario 1 when the correlation is 95%. Scenario 2 is more sensitive and even the ML model together with the 2step model starts to perform worse at lower levels of correlation compared to scenario 1 (85% correlation).

In scenario 3, it should be noted that in all cases the Heckman models perform worse than the SSLR model. Connecting the results of the predictions with 'test' with the results of the predictions with 'test_ws', it can be observed that when making predictions when there is selection bias ('test_ws') the Heckman models will predict with a certain average level of error because they are models that are biased by getting closer to the true relationship. However, these models are sensitive to changes in the correlation of variables, degree of selection bias, degree of censoring and sample size, which causes the error of their predictions to increase. This is not a problem because they are still closer to the true relationship compared to the SSLR model. But up to a certain limit where that error is already large

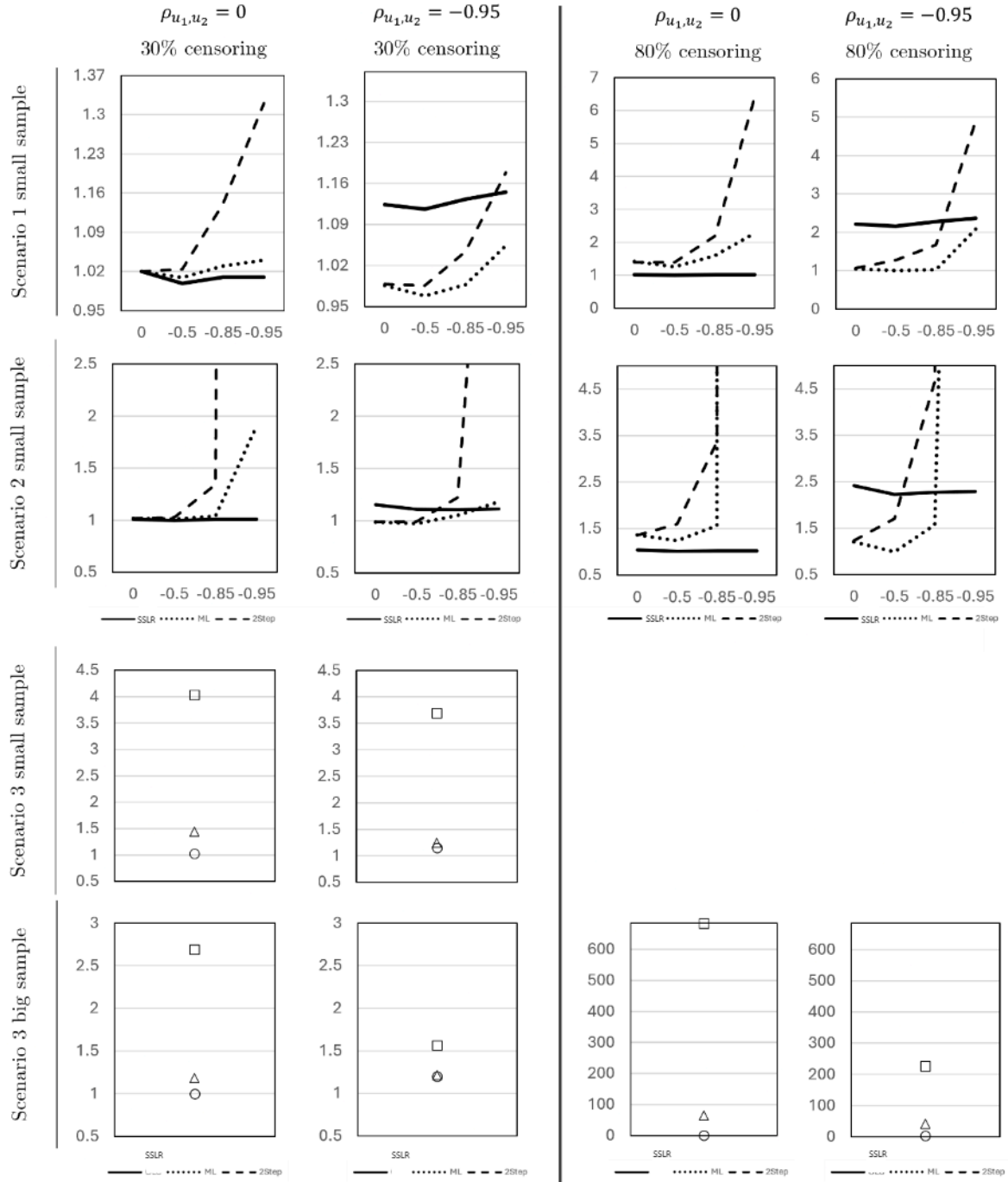


FIGURE 5.7: average MSE resulting from 100 prediction iterations in 'test' for cases where average MSE is bigger in Heckman models. y-axes represent different values corresponding to the average MSE.

enough to move away from the true relationship compared to the SSLR model. The less exclusion restriction, the more sensitive this limit is.

As a last remark, RMSE results can also be seen from the point of view of the MAE but they show the same, being MAE values slightly lower. See Appendix for detailed information of metrics. Additionally, prediction metrics from the True model can also be found in the Appendix. They are not shown in the figures for better clear visualization but when predicting in 'test' it is the model that predicts more accurately.

6. Ambulatory expenditures data

The data set from the 2001 Medical Expenditures used by **Cameron and Trivedi (2009, pp.541 -550)** in their book is one of the most widely used data sets when working with Heckman corrections. It used for instance by many of the references commented in this manuscript (**Marchenko & Genton ,2012** and **Zhelonkin, Genton & Ronchetti, 2016**).

This data set also works as a marketing related case where the specific ambulatory institution aims to predict the expenditures (lnambx) patients incur with them. This data counts on 3328 observations and sample selection bias is inferred because 16% of these observations (526) have zero expenditures (not recorded, like when someone buys a painkiller at a store without using insurance) and the rest a positive expenditure amount recorded (such as when a hospital visit takes place). The selection problem is defined as a set of variables included in both the selection and outcome equation, which are totchr (the number of chronic diseases), age, female (binary defining gender), educ (education status), blhisp (ethnicity) and ins (the insurance status). Additionally variable 'income' is used as the exclusion restriction.

The goal is to analyse how predictions coming from Heckman models differ from those coming from a linear regression considering only selected individuals (positive expenditure). When working with simulated data in the previous sections it was possible to check the prediction accuracy coming from predictions made on **"test"**. However, in a real case scenario the outcome variable from non-selected individuals is missing thus, predictions and their

comparison among models can only be done in a test set with selection (“**test_ws**”), that is, a testing set only considering individuals with positive expenditures recorded.

The same way as with the previous simulations, different scenarios and situations can be applied when predicting. The censoring rate and correlation of variables and errors cannot be changed as they are given by the problem itself, but the sample size and the exclusion restriction can be considered. These combinations are summarized in Table 6.1. The correlation between variables can be observed in the Appendix section. Underneath Table 6.1 a sample code example for predictions without exclusion restriction and small sample size is offered.

Models analysed	Estimators used	Sample size	Repetitions	Judgement criteria for predictions
With exclusion restriction	ML 2step	3328 (2997 for training & 331 for testing)	100	MSE (and decomposed MSE)
Without exclusion restriction	SSLR	832 (501 for training & 331 for testing)		RMSE MAE

TABLE 6.1: *Combinations for predictions with Ambulatory expenditures data*

```
R> for (i in 1:100) {
R> set.seed(i)
R> train_indices <- createDataPartition(data$age, p = 0.9, list = FALSE)
R> train <- data[train_indices, ]
R> test <- data[-train_indices, ]
R> train_ws <- train[train$dambexp == 1, ]
R> test_ws <- test[test$dambexp == 1, ]
R> selectEq <- dambexp ~ age + female + educ + blhisp + totchr + ins
  #To carry out predictions with selection restriction update code above adding variable "income"
  # selectEq <- dambexp ~ age + female + educ + blhisp + totchr + ins + income
```

```

R> outcomeEq <- lnambx ~ age + female + educ + blhisp + totchr + ins
R> ols <- lm(lnambx ~ age, data = train_ws)
R> ml <- selection(selectEq, outcomeEq, data = train, method = "ml")
R> 2step <- selection(selectEq, outcomeEq, data = train, method = "2step")
R> ...6

```

The variables used show a low correlation among them (see Appendix), so compared to the previous simulations, it would be similar to making predictions in Scenario 3 with low censoring degree and low correlation of variables. Therefore, it can be expected that the Heckman models make an error in their predictions similar to those of the SSLR model.

As it can be seen in Table 6.2, all models predict on average with an almost identical error, a little higher for the Heckman models but a very minimal difference. The models were trained taking the natural logarithm so, when transformed to the real units (last column) it can be seen that, compared to SSLR model, Heckman predictions deviate just in some cents.

As shown in Figure 6.1, although the difference is very small, when making predictions with a small sample, predicting with the 2step model can make higher errors compared to the ml model.

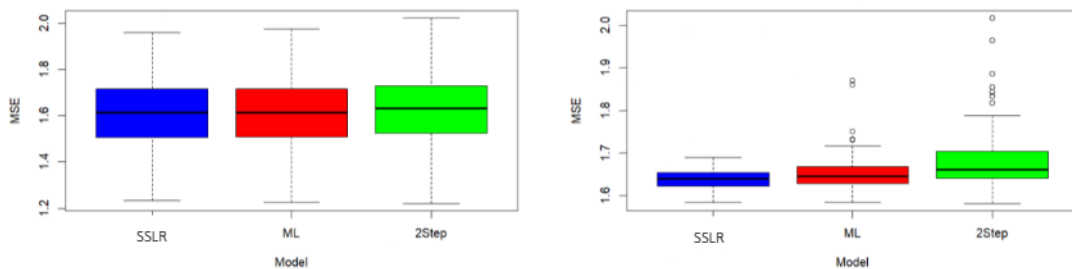


FIGURE 6.1: *Distribution of MSEs from 100 iterations with a big sample size (left) and a small sample size (right)*

⁶ The lines of code shown are extracts from the entire code. See Appendix for detailed information about the code. It is disposed as a loop to run train, predict and compute the metrics in one run.

Without Exclusion restriction Big sample size						
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Exp(RMSE)
SSLR	0.9946	1.264	1.602	1.601	0.0067	3.54
ML	0.995	1.266	1.606	1.602	0.0094	3.55
2Step	0.9996	1.274	1.628	1.61	0.024	3.58
Without Exclusion restriction Big sample size						
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Exp(RMSE)
SSLR	0.9946	1.264	1.602	1.601	0.0067	3.54
ML	0.995	1.266	1.606	1.602	0.0097	3.55
2Step	0.9992	1.274	1.626	1.609	0.023	3.58
With Exclusion restriction Big sample size						
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Exp(RMSE)
SSLR	1.007	1.28	1.638	1.634	0.0052	3.60
ML	1.01	1.286	1.653	1.638	0.0159	3.62
2Step	1.017	1.296	1.68	1.646	0.0346	3.65
Without Exclusion restriction Big sample size						
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	Exp(RMSE)
SSLR	1.007	1.28	1.638	1.634	0.0052	3.60
ML	1.01	1.286	1.655	1.639	0.0168	3.62
2Step	1.017	1.297	1.683	1.648	0.0356	3.66

TABLE 6.2: *summary of average metrics from different prediction iteration cases. From big sample to small sample and from exclusion restriction to without exclusion.*

These predictions consider models with several predictors. A more insightful view for the scope level of this thesis can be achieved by lowering the number as considered in the simulation section. This way predictions with high and low correlation between variables can also be analysed. Residuals metrics from these predictions are summarized in Table 6.3.

The last one represents scenario 1 (with exclusion restriction) being variables highly correlated and lower censoring. SSLR model predicts ambulatory expenditures on average with a mean error of 4\$ while ML with a mean error between 6 and 7\$. The significant values come from predicting with 2step model that makes on average and average error of 73\$ and even 106\$ if the sample is small. This situation was considered

in Figure 5.7 where for high correlation of variables SSLR actually has a better predictive accuracy than 2step while ML can still perform better than both.

In scenario 2 this was also the case but even with a lower correlation than 0.9 does SSLR predict more accurately. However this difference is not very visible when predicting with ambulatory data (second part of predictions in Table 6.3). The average error committed is similar to the one in scenario 1 but here 2step, though making a bigger average error, is significantly lower than the previous predictions.

Lastly, the last two predictions are done without exclusion restriction. Simulations showed that 2step makes an excessive bias and that SSLR model definitely predicts more accurately despite being a model trained to predict only selected individuals. When predicting ambulatory expenditures, it can be seen that while SSLR and ml make on average an error of a couple dollars, 2step gets to make on average an error of 280\$ when using income as a predictor. When predicting with age, this error is much lower. The difference is that the range of possible values for income variable is much greater than for age variable.

The same way as in the simulations, predicting without exclusion restriction when the sample is small gives an error and it is not possible.

Additionally, it is worth noting that when decomposing the average MSE, bias is predominant compared to the square bias of residuals when predicting with 2step. However, ml generates predictions with a lower average error, closer to the SSLR one but this difference is defined more for a slightly higher variance than bias. It would be interesting being able to predict in a sample with selected and non-selected individuals to understand if this defines ML as a better or worse predictive model than SSLR.

	Predictions Big Sample						Prediction Small Sample					
Scenario 3: selec_eq = age outcome_eq = age												
Model	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)						
SSLR	1.093	1.362	1.86	1.859	0.0072	3.9						
ML	1.314	1.598	2.76	1.944	0.8231	4.9						
2Step	1.984	2.316	5.447	2.121	3.3336	10.1						
Scenario 3: selec_eq = income outcome_eq = income												
Model	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)						
SSLR	1.132	1.404	1.976	1.975	0.0074	4.1						
ML	1.517	1.869	4.034	2.719	1.3245	6.5						
2Step	5.376	5.633	32.438	2.833	29.615	279.5						
Scenario 1: selec_eq = age2 outcome_eq = age cor _{age, age2} = 0.9												
Model	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)
SSLR	1.093	1.362	1.86	1.859	0.0072	3.9	1.1	1.372	1.882	1.877	0.0054	3.9
ML	1.359	1.7	3.046	2.736	0.32	5.5	1.591	1.966	4.446	3.938	0.5093	7.1
2Step	3.924	4.286	18.657	3.022	15.646	72.7	4.304	4.664	31.153	3.653	27.502	106.1
Scenario 2: selec_eq = age + age2 outcome_eq = age cor _{age, age2} = 0.9												
Model	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)	MAE	RMSE	MSE	Varian Residuals	Squa bias	Exp (RMSE)
SSLR	1.093	1.362	1.86	1.859	0.0072	3.9	1.1	1.372	1.882	1.877	0.0054	3.9
ML	1.237	1.519	2.475	2.027	0.4544	4.6	1.395	1.73	4.079	3.607	0.4734	5.6
2Step	1.81	2.127	5.048	2.088	2.9672	8.4	2.374	2.703	20	2.75	17.251	14.9

TABLE 6.3: summary of average metrics from different prediction iteration cases.

7. Conclusion

Sample selection bias is present in several situations, including in the field of Marketing. Making predictions with a non-random sample can lead to biased and erroneous results. The Heckman correction is one of the solutions that helps correct this selection bias.

In real cases, the predictive accuracy of the classic Heckman models can be compared with those of an SSLR (linear regression with selected observations) model through testing on a sample with selection bias since the selected individuals are the only ones who have information about the dependent variable of the outcome equation. When making these predictions and comparing the performance between models, the Heckman models will always make more errors on average compared to the SSLR model and as the selection bias increases, the larger this difference becomes. Factors that influence the criticism of Heckman's estimators also imply that the errors committed by

Heckman's models are greater compared to the SSLR model. Such as the degree of censorship, the correlation of variables and the exclusion restriction.

In principle, Heckman models predict better because when making predictions on a random sample they make less error than SSLR (except when there are really no indications of selection bias). However, there is a limit where the SSLR model, despite being a model trained taking into account a part of the population, performs better than Heckman models. This is mainly notable as there is less exclusion restriction and when the variables are highly correlated. Furthermore, the 2step model is more sensitive to these changes, making errors that are too high.

As previously mentioned, when predictions are made with a sample with selection bias, they will make a higher error on average using the Heckman models. If these models are used, to avoid high errors that cross the limit where a biased SSLR model is closer to the true relationship than a Heckman model, it is recommended to make predictions with a sufficiently large sample, with a low level of censoring where the variables are not so correlated and the fewer predictors of the outcome equation are present in the selection equation, the better.

Predictions with Ambulatory expenditures data showed that predictive accuracy, especially with 2step method becomes more sensible the bigger the range of values explanatory variables can take.

Additionally, introducing more predictors reduced the bias committed by Heckman models and became closer to SSLR model. This thesis should work as a reference for further studies in predictive accuracy where increase of number of explanatory variables and other scenarios such as other distribution assumptions of the error terms are considered, especially in marketing related fields to increase literature in this domain.

In a nutshell, what marketers need to have in mind is:

- Even if the sample is not chosen at random, information about the dependent variable of unobserved individuals must be collected in order to work with Heckman estimators.
- Predictions with the classical Heckman estimators will make more errors in their predictions compared to the bias regression estimator (SSLR) used before when

sample selection bias was not acknowledged, but they are actually predicting more accurately as they are correcting for sample selection bias and considering both unobserved and observed individuals in their predictions.

- A biased regression model only taking into account selected individuals can still be preferred to make better predictions if correlation among variables is very high, as well as censoring and degree of selection bias. In those cases Heckman estimators lose their predictive accuracy, especially 2step estimator thus, ml should be consider as a first option when predicting with Heckman correction models.

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9. Appendix

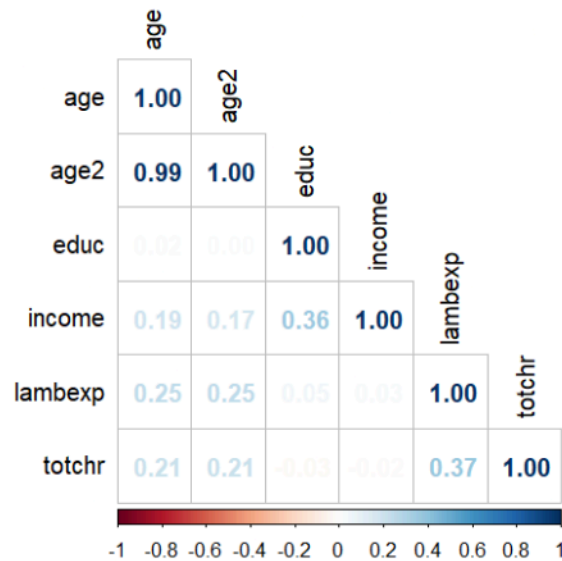


FIGURE 9.1: Correlation plot of numeric variables from Ambulatory expenditures data

The following figures show the metrics from all predictions. As predictions are repeated several times, the values are in average terms. The main division of tables is by sample size, degree of censoring and Scenarios. Tables in green represent Scenario 1, the ones in blue, Scenario 2 and the ones in red, Scenario 3. Within each table, the different predictions by correlation of variables, and correlation of errors. The values written in the ‘censoring’ are the values use in the code to generate selection at the specific desired censoring degree, low or high. The code for running simulations can be found on this GitHub repository.

<https://github.com/oicangii/Thesis/blob/4793ffdf229cf92500477e16030467e5d5c9acf2/R%20code%20thesis%20Ignacio%20Guzm%C3%A1n.Rmd>

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = -0.15, corr.errors = 0, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.798	0.981	0.963	0.961	0.0068	1.02	1.01	0.00531	-0.078	0.101	1.13	0.175
ML	0.802	0.985	0.972	0.961	0.01571	1.02	1.01	0.01453	0.075	0.116	1.13	0.219
2Step	0.801	0.984	0.97	0.961	0.01371	1.02	1.01	0.01249	0.073	0.115	1.13	0.218
TRUE	0.798	0.98	0.961	0.959	0.00581	1.01	1.01	0.0051	-0.052	0.088	1.07	0.15
censoring = -0.15, corr.errors = 0, corr.var = -0.5, N = 800												
SSLR	0.808	0.989	0.98	0.977	0.00689	0.999	0.997	0.00527	-0.042	0.102	1.04	0.18
ML	0.812	0.994	0.989	0.977	0.01642	1.009	0.997	0.01555	0.083	0.115	1.15	0.218
2Step	0.817	1.002	1.004	0.977	0.03145	1.024	0.997	0.03103	0.087	0.116	1.14	0.219
TRUE	0.808	0.988	0.977	0.976	0.0057	0.998	0.996	0.005	-0.062	0.091	1.09	0.15
censoring = -0.15, corr.errors = 0, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.795	0.975	0.951	0.949	0.00639	1.01	1	0.0056	-0.02	0.1	1.02	0.18
ML	0.805	0.987	0.976	0.951	0.02988	1.03	1.01	0.03035	0.076	0.115	1.15	0.218
2Step	0.845	1.038	1.082	0.958	0.12868	1.14	1.01	0.1346	0.083	0.116	1.14	0.219
TRUE	0.795	0.974	0.949	0.948	0.00557	1.01	1	0.0057	-0.048	0.091	1.07	0.15
censoring = -0.15, corr.errors = 0, corr.var = -0.95, N = 800												
SSLR	0.801	0.978	0.958	0.954	0.00826	1.01	1.01	0.00623	-0.016	0.099	0.99	0.179
ML	0.815	0.994	0.993	0.957	0.0409	1.04	1.01	0.0392	0.107	0.116	1.11	0.219
2Step	0.911	1.113	1.257	0.981	0.27994	1.32	1.03	0.29568	0.113	0.116	1.1	0.219
TRUE	0.801	0.977	0.956	0.954	0.00698	1.01	1.01	0.0059	-0.047	0.091	1.06	0.15
censoring = -0.15, corr.errors = -0.5, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.764	0.931	0.867	0.865	0.00565	1.043	0.995	0.05095	-0.242	0.095	1.03	0.163
ML	0.776	0.943	0.89	0.865	0.02886	1.005	0.995	0.01333	0.12	0.115	1.14	0.218
2Step	0.785	0.954	0.912	0.865	0.05048	1.003	0.995	0.01163	0.13	0.116	1.12	0.221
TRUE	0.785	0.954	0.911	0.865	0.05003	0.998	0.996	0.0052	-0.04	0.087	1.05	0.15
censoring = -0.15, corr.errors = -0.5, corr.var = -0.5, N = 800												
SSLR	0.761	0.927	0.861	0.858	0.0063	1.03	0.981	0.05278	-0.2	0.095	0.937	0.168
ML	0.774	0.937	0.879	0.859	0.0239	0.989	0.979	0.01282	0.144	0.115	1.063	0.216
2Step	0.8	0.968	0.94	0.86	0.0839	0.994	0.979	0.01864	0.158	0.116	1.036	0.219
TRUE	0.786	0.951	0.905	0.86	0.0493	0.981	0.979	0.0051	-0.051	0.09	1.076	0.15
censoring = -0.15, corr.errors = -0.5, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.753	0.919	0.846	0.844	0.00599	1.039	0.994	0.04801	-0.13	0.093	0.809	0.167
ML	0.769	0.934	0.874	0.845	0.03211	1.002	0.99	0.0145	0.147	0.116	1.087	0.22
2Step	0.834	1.012	1.03	0.855	0.17901	1.05	0.991	0.06217	0.157	0.117	1.066	0.221
TRUE	0.783	0.949	0.902	0.849	0.05701	0.991	0.989	0.0053	-0.039	0.09	1.059	0.154
censoring = -0.15, corr.errors = -0.5, corr.var = -0.95, N = 800												
SSLR	0.759	0.925	0.856	0.853	0.00712	1.048	0.997	0.05371	-0.112	0.092	0.762	0.167
ML	0.777	0.943	0.893	0.856	0.04083	1.03	0.994	0.0393	0.123	0.117	1.133	0.223
2Step	0.871	1.055	1.13	0.879	0.25482	1.125	1.001	0.12755	0.128	0.116	1.123	0.222
TRUE	0.787	0.952	0.907	0.86	0.05182	0.994	0.991	0.0058	-0.04	0.09	1.053	0.15
censoring = -0.15, corr.errors = -0.85, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.647	0.796	0.634	0.632	0.00451	1.124	0.979	0.14851	-0.404	0.081	1.04	0.14
ML	0.705	0.863	0.747	0.632	0.11771	0.986	0.979	0.01056	0.108	0.103	1.1	0.189
2Step	0.715	0.877	0.771	0.632	0.14192	0.988	0.979	0.0124	0.133	0.116	1.04	0.218
TRUE	0.719	0.881	0.778	0.632	0.14917	0.982	0.98	0.0054	-0.026	0.087	1.03	0.15
censoring = -0.15, corr.errors = -0.85, corr.var = -0.5, N = 800												
SSLR	0.646	0.792	0.629	0.627	0.00485	1.116	0.966	0.15333	-0.33	0.082	0.872	0.147
ML	0.713	0.869	0.757	0.63	0.13033	0.968	0.963	0.00821	0.027	0.105	1.208	0.197
2Step	0.761	0.93	0.87	0.632	0.24093	0.986	0.963	0.02627	0.075	0.115	1.098	0.217
TRUE	0.72	0.879	0.774	0.63	0.14775	0.967	0.964	0.0053	-0.039	0.089	1.063	0.15
censoring = -0.15, corr.errors = -0.85, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.639	0.783	0.614	0.613	0.00442	1.133	0.986	0.15008	-0.239	0.081	0.693	0.146
ML	0.717	0.875	0.768	0.622	0.14901	0.987	0.977	0.01299	0.068	0.11	1.12	0.203
2Step	0.814	0.991	0.992	0.635	0.36039	1.044	0.98	0.0672	0.095	0.115	1.055	0.216
TRUE	0.725	0.884	0.783	0.623	0.16364	0.979	0.977	0.00486	-0.029	0.089	1.047	0.152
censoring = -0.15, corr.errors = -0.85, corr.var = -0.95, N = 800												
SSLR	0.639	0.784	0.615	0.613	0.00482	1.145	0.988	0.16097	-0.223	0.08	0.649	0.145
ML	0.775	0.942	0.911	0.639	0.27491	1.054	0.983	0.07384	0.096	Inf	1.083	Inf
2Step	0.901	1.088	1.22	0.66	0.56284	1.178	0.991	0.18999	0.113	0.115	1.04	0.217
TRUE	0.721	0.88	0.775	0.625	0.15274	0.979	0.977	0.0057	-0.031	0.089	1.043	0.152

FIGURE 9.2: Predictions Scenario 1, small sample size, low censoring degree ($\sim 25\%$)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = 1.35, corr.errors = 0, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.739	0.914	0.839	0.831	0.0219	1.02	1.02	0.00728	-0.135	0.172	1.201	0.299
ML	0.889	1.093	1.237	0.831	0.4196	1.42	1.02	0.4027	-1.32	0.14	0.919	0.229
2Step	0.877	1.079	1.203	0.831	0.3856	1.39	1.02	0.37144	-1.322	0.139	0.921	0.228
TRUE	0.732	0.903	0.818	0.821	0.0107	1.01	1.01	0.0051	-0.052	0.088	1.066	0.152
censoring = 1.35, corr.errors = 0, corr.var = -0.5, N = 800												
SSLR	0.739	0.91	0.831	0.827	0.0184	1.019	1.012	0.0101	-0.212	0.172	1.507	0.323
ML	0.818	1	1.015	0.829	0.2006	1.253	1.018	0.2385	-1.373	0.141	0.991	0.23
2Step	0.867	1.05	1.143	0.829	0.3278	1.398	1.02	0.3814	-1.37	0.142	0.985	0.231
TRUE	0.737	0.91	0.832	0.834	0.0118	0.998	0.996	0.005	-0.062	0.091	1.086	0.15
censoring = 1.35, corr.errors = 0, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.742	0.913	0.837	0.83	0.0213	1.02	1.01	0.01092	-0.152	0.166	1.36	0.336
ML	0.926	1.124	1.295	0.839	0.4698	1.6	1.04	0.56802	-1.413	0.143	1.05	0.231
2Step	1.091	1.293	1.799	0.848	0.9653	2.19	1.05	1.14	-1.409	0.143	1.04	0.232
TRUE	0.733	0.906	0.824	0.826	0.0116	1.01	1	0.0057	-0.048	0.091	1.07	0.15
censoring = 1.35, corr.errors = 0, corr.var = -0.95, N = 800												
SSLR	0.722	0.89	0.795	0.785	0.0231	1.02	1.02	0.01071	-0.139	0.158	1.334	0.319
ML	1.05	1.25	1.839	0.801	1.0513	2.27	1.05	1.21735	-1.319	0.137	0.938	0.226
2Step	1.954	2.139	5.468	0.868	4.6141	6.39	1.15	5.24631	-1.31	0.138	0.922	0.227
TRUE	0.713	0.883	0.783	0.783	0.013	1.01	1.01	0.0059	-0.047	0.091	1.06	0.15
censoring = 1.35, corr.errors = -0.5, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.661	0.831	0.694	0.689	0.0165	1.411	0.998	0.4169	-0.739	0.155	1.18	0.275
ML	0.732	0.903	0.827	0.689	0.1497	1.278	0.997	0.2838	-1.425	0.144	1.04	0.233
2Step	0.749	0.922	0.866	0.689	0.1892	1.264	0.997	0.27031	-1.423	0.144	1.03	0.234
TRUE	0.857	1.038	1.081	0.681	0.412	0.998	0.996	0.00528	-0.04	0.087	1.05	0.15
censoring = 1.35, corr.errors = -0.5, corr.var = -0.5, N = 800												
SSLR	0.637	0.795	0.635	0.631	0.015	1.397	0.982	0.41842	-0.778	0.152	1.241	0.287
ML	0.831	1.002	1.022	0.633	0.4	1.046	0.987	0.06326	-1.389	0.14	1.006	0.226
2Step	1.056	1.229	1.602	0.635	0.979	1.248	0.99	0.26128	-1.373	0.142	0.978	0.231
TRUE	0.841	1.018	1.04	0.625	0.426	0.981	0.979	0.00519	-0.051	0.09	1.076	0.15
censoring = 1.35, corr.errors = -0.5, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.625	0.788	0.624	0.618	0.0175	1.416	0.991	0.42791	-0.673	0.148	1.01	0.301
ML	0.864	1.045	1.112	0.626	0.4969	1.05	0.998	0.05531	-1.431	0.143	1.06	0.231
2Step	1.477	1.648	2.935	0.649	2.2975	2.012	1.023	0.99218	-1.412	0.144	1.03	0.234
TRUE	0.841	1.022	1.049	0.607	0.453	0.991	0.989	0.0053	-0.039	0.09	1.06	0.15
censoring = 1.35, corr.errors = -0.5, corr.var = -0.95, N = 800												
SSLR	0.648	0.813	0.664	0.654	0.0215	1.409	0.994	0.41817	-0.599	0.147	0.892	0.308
ML	0.927	1.108	1.366	0.669	0.7089	1.21	1.002	0.21132	-1.438	0.144	1.087	0.233
2Step	2.176	2.339	6.091	0.771	5.3334	4.558	1.099	3.46254	-1.423	0.144	1.062	0.233
TRUE	0.847	1.027	1.06	0.646	0.425	0.994	0.991	0.00588	-0.04	0.09	1.053	0.15
censoring = 1.35, corr.errors = -0.85, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.501	0.641	0.414	0.41	0.0111	2.21	0.983	1.22973	-1.24	0.123	1.25	0.213
ML	0.972	1.12	1.276	0.409	0.8743	1.049	0.981	0.07145	-1.43	0.135	1.08	0.216
2Step	1.188	1.325	1.822	0.409	1.4197	1.069	0.982	0.09062	-1.403	0.143	1.03	0.233
TRUE	1.133	1.275	1.63	0.407	1.2305	0.982	0.98	0.00549	-0.026	0.087	1.03	0.15
censoring = 1.35, corr.errors = -0.85, corr.var = -0.5, N = 800												
SSLR	0.492	0.626	0.394	0.39	0.011	2.157	0.964	1.19594	-1.114	0.116	1.03	0.225
ML	1.051	1.201	1.465	0.396	1.076	0.998	0.969	0.03198	-1.479	0.132	1.19	0.21
2Step	1.472	1.598	2.664	0.398	2.273	1.269	0.971	0.30118	-1.424	0.144	1.1	0.232
TRUE	1.105	1.247	1.562	0.384	1.184	0.967	0.964	0.0053	-0.039	0.089	1.06	0.15
censoring = 1.35, corr.errors = -0.85, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.482	0.615	0.381	0.377	0.0102	2.278	0.998	1.28337	-0.906	0.111	0.532	0.233
ML	0.978	1.131	1.3	0.396	0.9111	1.027	0.978	0.05233	-1.503	0.141	1.232	0.225
2Step	1.725	1.846	3.568	0.433	3.1423	1.675	0.992	0.68634	-1.456	0.145	1.154	0.234
TRUE	1.105	1.249	1.566	0.389	1.1831	0.979	0.977	0.00486	-0.029	0.089	1.047	0.15
censoring = 1.35, corr.errors = -0.85, corr.var = -0.95, N = 800												
OLS	0.485	0.617	0.383	0.379	0.0103	2.369	1.006	1.36624	-0.88	0.109	0.424	0.23
ML	1.307	1.454	2.94	0.44	2.5074	2.08	1.001	1.0827	-1.465	Inf	1.172	Inf
2Step	2.768	2.874	8.768	0.584	8.194	4.904	1.08	3.82806	-1.432	0.144	1.117	0.233
TRUE	1.119	1.263	1.601	0.402	1.2056	0.979	0.977	0.0057	-0.031	0.089	1.043	0.15

FIGURE 9.3: Predictions Scenario 1, small sample size, high censoring degree (~80%)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = 0.15, corr.errors = 0, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.802	1	1.01	1.01	0.00509	0.998	0.997	0.00396	-0.013	0.057	0.962	0.098
ML	0.803	1.01	1.01	1.01	0.00671	1	0.997	0.00575	-0.156	0.056	1.018	0.102
2Step	0.803	1.01	1.01	1.01	0.00688	1	0.997	0.00591	-0.156	0.056	1.018	0.102
TRUE	0.802	1	1.01	1.01	0.00474	0.997	0.997	0.0036	-0.034	0.045	1.033	0.07
censoring = 0.15, corr.errors = 0, corr.var = -0.5, N = 2300												
SSLR	0.799	0.999	1	1	0.00493	0.984	0.984	0.00322	-0.022	0.056	0.994	0.1
ML	0.8	1.001	1	1	0.00788	0.987	0.984	0.00634	-0.156	0.057	0.976	0.102
2Step	0.8	1	1	1	0.00761	0.987	0.984	0.00603	-0.156	0.057	0.976	0.102
TRUE	0.799	0.999	1	1	0.00463	0.984	0.984	0.0032	0.01	0.045	0.949	0.07
censoring = 0.15, corr.errors = 0, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.802	1	1.01	1.01	0.00443	0.988	0.988	0.00307	-0.02	0.055	0.981	0.1
ML	0.808	1.01	1.02	1.01	0.01865	1.004	0.989	0.0189	-0.168	0.056	1.006	0.102
2Step	0.805	1.01	1.02	1.01	0.01274	0.998	0.989	0.01295	-0.168	0.056	1.007	0.102
TRUE	0.802	1	1.01	1.01	0.00418	0.988	0.988	0.0029	0.016	0.045	0.936	0.078
censoring = 0.15, corr.errors = 0, corr.var = -0.95, N = 2300												
SSLR	0.803	1.01	1.02	1.02	0.00606	1	1	0.00381	-0.027	0.054	1	0.101
ML	0.813	1.02	1.04	1.02	0.03338	1.04	1	0.03675	-0.172	0.056	1.019	0.102
2Step	0.812	1.02	1.04	1.02	0.02956	1.03	1	0.03141	-0.172	0.056	1.021	0.102
TRUE	0.803	1.01	1.02	1.02	0.00616	1	1	0.00382	0.017	0.045	0.937	0.07
censoring = 0.15, corr.errors = -0.5, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.765	0.957	0.919	0.919	0.00422	1.06	1.01	0.0524	-0.207	0.05	0.951	0.087
ML	0.772	0.971	0.945	0.919	0.03089	1.02	1.01	0.00742	0.149	0.058	1.007	0.108
2Step	0.776	0.977	0.956	0.919	0.04146	1.02	1.01	0.00578	0.152	0.058	1	0.108
TRUE	0.779	0.981	0.964	0.919	0.04897	1.01	1.01	0.0037	-0.03	0.045	1.03	0.07
censoring = 0.15, corr.errors = -0.5, corr.var = -0.5, N = 2300												
SSLR	0.759	0.949	0.902	0.902	0.00408	1.05	1	0.05349	-0.194	0.05	0.911	0.088
ML	0.768	0.964	0.932	0.903	0.03305	1.01	1	0.00752	0.15	0.058	0.97	0.107
2Step	0.769	0.966	0.934	0.903	0.03569	1.01	1	0.00718	0.152	0.058	0.967	0.108
TRUE	0.776	0.975	0.952	0.902	0.054	1	1	0.0036	-0.005	0.046	0.981	0.07
censoring = 0.15, corr.errors = -0.5, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.767	0.956	0.917	0.917	0.00399	1.06	1.01	0.05294	-0.152	0.049	0.828	0.088
ML	0.771	0.965	0.933	0.917	0.02047	1.02	1.01	0.01496	0.162	0.058	0.958	0.108
2Step	0.772	0.968	0.939	0.917	0.02616	1.01	1.01	0.01162	0.163	0.058	0.955	0.108
TRUE	0.781	0.981	0.964	0.917	0.05117	1	1.01	0.0029	0.002	0.046	0.969	0.07
censoring = 0.15, corr.errors = -0.5, corr.var = -0.95, N = 2300												
SSLR	0.763	0.958	0.92	0.92	0.00477	1.07	1.02	0.0524	-0.141	0.049	0.804	0.088
ML	0.767	0.966	0.935	0.921	0.01875	1.05	1.02	0.02627	0.148	0.058	0.991	0.108
2Step	0.775	0.978	0.958	0.922	0.03982	1.04	1.02	0.02074	0.148	0.058	0.989	0.108
TRUE	0.781	0.987	0.977	0.922	0.05867	1.02	1.02	0.0041	0.003	0.046	0.968	0.07
censoring = 0.15, corr.errors = -0.85, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.665	0.83	0.691	0.691	0.00298	1.19	1.03	0.15812				
ML	0.717	0.917	0.844	0.691	0.1559	1.03	1.03	0.00398	1.19	1.03	0.15812	
2Step	0.708	0.904	0.819	0.691	0.13163	1.03	1.03	0.00656	1.03	1.03	0.00398	
TRUE	0.717	0.917	0.843	0.691	0.15542	1.03	1.03	0.0038	1.03	1.03	0.00656	
censoring = 0.15, corr.errors = -0.85, corr.var = -0.5, N = 2300												
SSLR	0.664	0.829	0.688	0.688	0.00331	1.18	1.02	0.155	-0.307	0.043	0.817	0.077
ML	0.725	0.924	0.856	0.693	0.16649	1.02	1.02	0.00433	0.138	0.05	0.982	0.091
2Step	0.709	0.902	0.816	0.692	0.12783	1.02	1.02	0.0059	0.135	0.058	0.981	0.108
TRUE	0.719	0.916	0.841	0.692	0.1525	1.02	1.02	0.004	-0.018	0.046	1.013	0.07
censoring = 0.15, corr.errors = -0.85, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.667	0.831	0.691	0.691	0.00334	1.19	1.03	0.15884	-0.241	0.042	0.67	0.077
ML	0.725	0.925	0.857	0.701	0.15954	1.02	1.02	0.00354	0.128	0.053	1	0.098
2Step	0.704	0.896	0.805	0.697	0.11111	1.03	1.02	0.00872	0.125	0.058	1	0.108
TRUE	0.72	0.919	0.846	0.699	0.15052	1.02	1.02	0.0031	-0.012	0.046	1	0.07
censoring = 0.15, corr.errors = -0.85, corr.var = -0.95, N = 2300												
SSLR	0.661	0.825	0.682	0.681	0.00344	1.21	1.05	0.16101	-0.234	0.042	0.646	0.076
ML	0.726	0.928	0.864	0.695	0.17177	1.04	1.04	0.00464	0.131	0.055	0.978	0.101
2Step	0.708	0.903	0.818	0.691	0.12988	1.05	1.04	0.01501	0.126	0.058	0.986	0.107
TRUE	0.722	0.923	0.854	0.692	0.16552	1.04	1.04	0.0044	-0.01	0.046	0.999	0.07

FIGURE 9.4: Predictions Scenario 1, big sample size, low censoring degree (~25%)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = 1.35, corr.errors = 0, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.809	1.01	1.03	1.03	0.019	1.007	0.999	0.01106	-0.181	0.101	1.17	0.173
ML	0.831	1.05	1.11	1.03	0.0974	1.124	0.999	0.12791	-1.382	0.071	1.12	0.114
2Step	0.829	1.05	1.1	1.03	0.0929	1.118	0.999	0.12213	-1.383	0.07	1.12	0.114
TRUE	0.813	1.01	1.03	1.02	0.0199	0.997	0.997	0.00363	-0.034	0.045	1.03	0.07
censoring = 1.35, corr.errors = 0, corr.var = -0.5, N = 2300												
SSLR	0.79	0.99	0.986	0.984	0.0177	0.995	0.99	0.00838	-0.197	0.089	1.213	0.171
ML	0.797	1.004	1.015	0.984	0.047	1.052	0.989	0.0666	-1.385	0.071	1.108	0.115
2Step	0.796	1.004	1.015	0.984	0.0464	1.05	0.989	0.0649	-1.385	0.071	1.108	0.115
TRUE	0.8	0.993	0.992	0.985	0.0227	0.984	0.984	0.0032	0.01	0.045	0.949	0.07
censoring = 1.35, corr.errors = 0, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.8	0.997	1	1	0.0138	0.996	0.992	0.00646	-0.16	0.086	1.165	0.176
ML	0.817	1.014	1.04	1	0.0496	1.03	0.995	0.03811	-1.398	0.071	1.143	0.115
2Step	0.824	1.022	1.05	1	0.0658	1.045	0.996	0.05187	-1.398	0.071	1.142	0.115
TRUE	0.809	1.001	1.01	1	0.0197	0.988	0.988	0.00296	0.016	0.045	0.936	0.07
censoring = 1.35, corr.errors = 0, corr.var = -0.95, N = 2300												
SSLR	0.803	1.01	1.03	1.03	0.0163	1.01	1.01	0.00551	-0.168	0.084	1.207	0.177
ML	0.866	1.07	1.16	1.03	0.1515	1.16	1.02	0.13902	-1.407	0.071	1.152	0.115
2Step	0.932	1.15	1.34	1.03	0.3217	1.33	1.03	0.30307	-1.407	0.071	1.151	0.115
TRUE	0.812	1.01	1.03	1.03	0.0229	1	1	0.0038	0.017	0.045	0.937	0.07
censoring = 1.35, corr.errors = -0.5, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.711	0.888	0.793	0.79	0.0144	1.53	1.01	0.51503	-0.738	0.088	1.02	0.15
ML	0.805	0.997	1.001	0.79	0.2224	1.09	1.01	0.07724	-1.345	0.07	1.1	0.113
2Step	0.813	1.007	1.021	0.79	0.2427	1.08	1.01	0.0675	-1.344	0.07	1.1	0.113
TRUE	0.916	1.126	1.275	0.788	0.4991	1.01	1.01	0.00375	-0.03	0.045	1.03	0.07
censoring = 1.35, corr.errors = -0.5, corr.var = -0.5, N = 2300												
SSLR	0.699	0.879	0.777	0.775	0.0138	1.53	1	0.53388	-0.684	0.078	0.883	0.149
ML	0.807	0.998	1.003	0.778	0.237	1.07	1	0.07293	-1.391	0.071	1.164	0.115
2Step	0.817	1.01	1.027	0.778	0.2608	1.06	1	0.06189	-1.39	0.071	1.162	0.115
TRUE	0.93	1.137	1.3	0.774	0.5384	1	1	0.0036	-0.005	0.046	0.981	0.07
censoring = 1.35, corr.errors = -0.5, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.699	0.875	0.77	0.768	0.0129	1.55	1.01	0.54125	-0.626	0.075	0.756	0.153
ML	0.794	0.983	0.978	0.773	0.2159	1.11	1.01	0.10757	-1.376	0.07	1.142	0.114
2Step	0.806	0.997	1.007	0.774	0.2454	1.09	1.01	0.09205	-1.376	0.07	1.142	0.114
TRUE	0.918	1.125	1.274	0.769	0.5167	1	1.01	0.0029	0.002	0.046	0.969	0.07
censoring = 1.35, corr.errors = -0.5, corr.var = -0.95, N = 2300												
SSLR	0.702	0.877	0.774	0.773	0.0131	1.57	1.03	0.54326	-0.599	0.074	0.693	0.153
ML	0.769	0.956	0.923	0.777	0.158	1.23	1.02	0.21218	-1.366	0.07	1.131	0.114
2Step	0.806	1	1.018	0.78	0.2498	1.21	1.02	0.19182	-1.366	0.07	1.13	0.114
TRUE	0.92	1.133	1.292	0.776	0.5284	1.02	1.02	0.0041	0.003	0.046	0.968	0.07
censoring = 1.35, corr.errors = -0.85, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.557	0.697	0.488	0.486	0.00928	2.32	1.03	1.29189	-1.197	0.068	1.1	0.115
ML	1	1.179	1.399	0.486	0.92045	1.06	1.03	0.03481	-1.349	0.066	1.1	0.108
2Step	1.019	1.197	1.443	0.486	0.96464	1.06	1.03	0.03164	-1.349	0.07	1.1	0.114
TRUE	1.141	1.313	1.732	0.485	1.25444	1.03	1.03	0.0038	-0.024	0.046	1.03	0.07
censoring = 1.35, corr.errors = -0.85, corr.var = -0.5, N = 2300												
SSLR	0.531	0.671	0.453	0.453	0.00762	2.36	1.03	1.33861	-1.024	0.06	0.719	0.115
ML	0.925	1.106	1.229	0.459	0.77705	1.09	1.02	0.07495	-1.379	0.07	1.135	0.113
2Step	0.863	1.043	1.095	0.458	0.64331	1.14	1.02	0.12498	-1.392	0.071	1.157	0.115
TRUE	1.15	1.319	1.746	0.46	1.29285	1.02	1.02	0.004	-0.018	0.046	1.013	0.07
censoring = 1.35, corr.errors = -0.85, corr.var = -0.85, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.538	0.673	0.457	0.456	0.00745	2.43	1.04	1.38945	-0.94	0.058	0.506	0.119
ML	0.997	1.182	1.404	0.478	0.93325	1.05	1.02	0.03393	-1.391	0.069	1.158	0.112
2Step	0.967	1.151	1.34	0.475	0.87166	1.08	1.02	0.05883	-1.397	0.071	1.169	0.115
TRUE	1.147	1.32	1.747	0.472	1.282	1.02	1.02	0.0031	-0.012	0.046	1.002	0.07
censoring = 1.35, corr.errors = -0.85, corr.var = -0.95, N = 2300												
SSLR	0.537	0.673	0.456	0.454	0.00818	2.47	1.07	1.40932	-0.908	0.057	0.421	0.119
ML	1.003	1.189	1.421	0.483	0.94529	1.07	1.04	0.0305	-1.375	0.07	1.141	0.113
2Step	1.1	1.278	1.664	0.488	1.18275	1.09	1.04	0.05154	-1.376	0.07	1.144	0.114
TRUE	1.144	1.318	1.744	0.477	1.27454	1.04	1.04	0.0044	-0.01	0.046	0.999	0.07

FIGURE 9.5: Predictions Scenario 1, big sample size, high censoring degree (~80%)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = 0.35, corr.errors = 0, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.81	0.992	0.985	0.983	0.00647	1.01	1.01	0.00532	-0.03	0.11	1.02	0.179
ML	0.813	0.995	0.991	0.984	0.01147	1.02	1.01	0.01077	-0.473	0.16	1.13	0.217
2Step	0.813	0.996	0.993	0.984	0.01339	1.02	1.01	0.0127	-0.473	0.16	1.13	0.216
TRUE	0.809	0.991	0.983	0.982	0.00564	1.01	1.01	0.0051	-0.052	0.088	1.07	0.152
censoring = 0.35, corr.errors = 0, corr.var = -0.5, N = 800												
OLS	0.802	0.983	0.967	0.965	0.00666	0.998	0.996	0.00518	-0.039	0.107	1.05	0.178
ML	0.805	0.988	0.977	0.965	0.01582	1.011	0.997	0.0169	-0.373	0.214	1.06	0.248
2Step	0.81	0.995	0.991	0.966	0.02974	1.027	0.998	0.03219	-0.367	0.215	1.06	0.249
TRUE	0.802	0.982	0.965	0.964	0.00572	0.998	0.996	0.005	-0.062	0.091	1.09	0.15
censoring = 0.35, corr.errors = 0, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.8	0.985	0.971	0.969	0.00656	1.01	1.01	0.00634	0.014	0.105	0.982	0.178
ML	0.809	0.997	0.996	0.969	0.03149	1.04	1.01	0.03505	-0.137	0.366	0.814	0.381
2Step	0.912	1.12	1.284	0.971	0.31695	1.34	1.01	0.33375	-0.105	0.369	0.785	0.383
TRUE	0.8	0.984	0.97	0.968	0.00583	1.01	1	0.00575	-0.048	0.091	1.068	0.15
censoring = 0.35, corr.errors = 0, corr.var = -0.95, N = 800												
OLS	0.798	0.981	0.964	0.96	0.00796	1.01	1.01	0.00635	-0.011	0.104	1.009	0.176
ML	0.958	1.164	1.824	1.426	0.40377	1.91	1.49	0.42536	-0.059	Inf	0.769	Inf
2Step	2.28	2.48	54.737	0.981	53.76062	54.75	1.03	53.72571	-0.035	0.636	0.745	0.643
TRUE	0.798	0.98	0.962	0.96	0.00675	1.01	1.01	0.0059	-0.047	0.091	1.06	0.15
censoring = 0.35, corr.errors = -0.5, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.758	0.921	0.85	0.848	0.00593	1.057	1	0.06019	-0.382	0.101	1.27	0.165
ML	0.774	0.937	0.879	0.85	0.0326	1.006	0.996	0.01291	-0.453	0.159	1.07	0.217
2Step	0.788	0.955	0.914	0.853	0.06447	1.004	0.996	0.01083	-0.434	0.16	1.06	0.216
TRUE	0.784	0.949	0.902	0.851	0.05499	0.998	0.996	0.00528	-0.04	0.087	1.05	0.15
censoring = 0.35, corr.errors = -0.5, corr.var = -0.5, N = 800												
OLS	0.754	0.92	0.847	0.845	0.00579	1.033	0.981	0.05506	-0.356	0.101	1.233	0.169
ML	0.768	0.932	0.871	0.846	0.0289	0.989	0.98	0.01236	-0.389	0.21	1.005	0.243
2Step	0.802	0.976	0.955	0.848	0.11107	1.005	0.981	0.0279	-0.356	0.212	0.984	0.243
TRUE	0.778	0.945	0.894	0.846	0.0524	0.981	0.979	0.00519	-0.051	0.09	1.076	0.15
censoring = 0.35, corr.errors = -0.5, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.748	0.913	0.835	0.832	0.00648	1.03	0.989	0.04392	-0.28	0.098	1.124	0.167
ML	0.763	0.928	0.862	0.832	0.03403	1.001	0.989	0.0147	-0.302	0.368	0.942	0.382
2Step	0.853	1.04	1.095	0.833	0.26618	1.1	0.99	0.11396	-0.243	0.372	0.888	0.386
TRUE	0.772	0.941	0.886	0.832	0.05784	0.991	0.989	0.0053	-0.039	0.09	1.059	0.15
censoring = 0.35, corr.errors = -0.5, corr.var = -0.95, N = 800												
OLS	0.754	0.917	0.842	0.839	0.00692	1.037	0.991	0.0501	-0.25	0.097	1.05	0.166
ML	0.826	1	1.075	0.901	0.17844	1.22	1.054	0.17015	-0.013	Inf	0.652	Inf
2Step	1.594	1.782	10.84	0.847	9.99618	10.684	1.001	9.68658	-0.015	0.632	0.651	0.638
TRUE	0.776	0.942	0.888	0.838	0.05395	0.994	0.991	0.0058	-0.04	0.09	1.053	0.15
censoring = 0.35, corr.errors = -0.85, corr.var = 0, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.655	0.802	0.644	0.641	0.00507	1.153	0.996	0.16043	-0.638	0.089	1.48	0.145
ML	0.716	0.87	0.759	0.652	0.11005	0.99	0.981	0.01304	-0.498	0.15	1.14	0.212
2Step	0.724	0.881	0.779	0.655	0.12671	0.989	0.981	0.01186	-0.473	0.16	1.11	0.216
TRUE	0.734	0.893	0.8	0.658	0.14456	0.982	0.98	0.0054	-0.026	0.087	1.03	0.15
censoring = 0.35, corr.errors = -0.85, corr.var = -0.5, N = 800												
OLS	0.645	0.79	0.626	0.624	0.00452	1.11	0.97	0.14311	-0.561	0.086	1.35	0.144
ML	0.709	0.861	0.744	0.628	0.11847	0.971	0.966	0.00882	-0.426	0.194	1.1	0.235
2Step	0.757	0.925	0.862	0.632	0.23227	0.993	0.966	0.03019	-0.367	0.213	1.06	0.246
TRUE	0.718	0.874	0.766	0.631	0.13809	0.967	0.964	0.0053	-0.039	0.089	1.06	0.15
censoring = 0.35, corr.errors = -0.85, corr.var = -0.85, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.639	0.781	0.61	0.609	0.00429	1.105	0.977	0.13129	-0.448	0.083	1.153	0.141
ML	0.796	0.972	0.96	0.611	0.35195	1.053	0.979	0.07814	-0.282	Inf	0.947	Inf
2Step	0.928	1.12	1.297	0.611	0.68883	1.223	0.978	0.24752	-0.178	0.372	0.844	0.386
TRUE	0.712	0.87	0.759	0.61	0.15199	0.979	0.977	0.0048	-0.029	0.089	1.047	0.15
censoring = 0.35, corr.errors = -0.85, corr.var = -0.95, N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
OLS	0.645	0.788	0.622	0.621	0.00447	1.113	0.976	0.14033	-0.419	0.083	1.086	0.141
ML	0.781	0.95	0.943	0.646	0.3	1.181	1.002	0.18315	-0.037	Inf	0.695	Inf
2Step	1.717	1.884	6.702	0.631	6.07329	6.5	0.989	5.51423	0.06	0.635	0.6	0.642
TRUE	0.711	0.869	0.757	0.62	0.13948	0.979	0.977	0.0057	-0.031	0.089	1.043	0.15

FIGURE 9.6: Predictions Scenario 2, small sample size, low censoring degree ($\sim 25\%$)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
- generate_normal(seed = 22, censoring = 1.9, corr.errors = 0, corr.var = 0, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals				Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.772	0.947	0.901	0.891	0.0251	1.04	1.03	0.01473	-0.287	0.222	1.388	0.335
ML	0.865	1.065	1.16	0.901	0.2739	1.37	1.04	0.3279	-1.953	0.203	0.952	0.233
2Step	0.863	1.064	1.147	0.901	0.2618	1.36	1.04	0.3165	-1.955	0.203	0.955	0.233
TRUE	0.769	0.937	0.881	0.883	0.013	1.01	1.01	0.00511	-0.052	0.088	1.066	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = 0, corr.var = -0.5, N = 800)												
SSLR	0.729	0.905	0.824	0.82	0.019	1.013	1.01	0.00672	-0.219	0.216	1.44	0.342
ML	0.817	1.001	1.015	0.819	0.211	1.239	1.007	0.23596	-1.772	0.245	0.885	0.263
2Step	0.937	1.127	1.348	0.821	0.5417	1.597	1.006	0.59434	-1.765	0.245	0.881	0.263
TRUE	0.725	0.898	0.811	0.814	0.0119	0.998	0.996	0.005	-0.062	0.091	1.086	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = 0, corr.var = -0.85, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.727	0.899	0.813	0.804	0.0245	1.02	1.02	0.00922	-0.176	0.213	1.37	0.352
ML	0.945	1.142	1.339	0.805	0.5499	1.56	1.01	0.54709	-2.028	0.408	1.1	0.414
2Step	1.426	1.619	3.131	0.807	2.3396	3.37	1.01	2.35915	-1.987	0.415	1.06	0.419
TRUE	0.718	0.888	0.794	0.797	0.0123	1.01	1	0.00575	-0.048	0.091	1.07	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = 0, corr.var = -0.95, N = 800)												
SSLR	0.747	0.921	8.52E-01	0.839	2.93E-02	1.02	1.02	9.27E-03	-0.194	0.21	1.37	0.356
ML	10.437	10.743	6.11E+03	178.769	5.93E+03	6193.07	180.58	6.01E+03	-2.151	Inf	1.22	Inf
2Step	23.342	23.485	3.17E+04	11.456	3.17E+04	31790.84	11.73	3.18E+04	-1.987	0.691	1.06	0.693
TRUE	0.74	0.911	8.34E-01	0.835	1.49E-02	1.01	1.01	5.94E-03	-0.047	0.091	1.06	0.155
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.5, corr.var = 0, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.657	0.821	0.677	0.671	0.0176	1.517	1.022	0.49911	-0.996	0.195	1.57	0.291
ML	0.722	0.887	0.796	0.672	0.1354	1.253	1.011	0.24609	-1.995	0.205	1.01	0.235
2Step	0.767	0.935	0.897	0.674	0.2348	1.238	1.009	0.2323	-1.991	0.205	1.01	0.235
TRUE	0.851	1.031	1.066	0.677	0.4014	0.998	0.996	0.00528	-0.04	0.087	1.05	0.151
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.5, corr.var = -0.5, N = 800)												
SSLR	0.659	0.829	0.692	0.685	0.0198	1.5	0.999	0.50379	-0.993	0.195	1.54	0.316
ML	0.919	1.106	1.264	0.688	0.5891	1.089	0.989	0.10245	-1.762	0.242	0.777	0.262
2Step	1.207	1.391	2.136	0.691	1.4577	1.469	0.987	0.48539	-1.743	0.246	0.766	0.263
TRUE	0.871	1.061	1.135	0.69	0.4578	0.981	0.979	0.00519	-0.051	0.09	1.076	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.5, corr.var = -0.85, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.673	0.844	0.717	0.708	0.0221	1.467	0.995	0.4757	-0.836	0.194	1.258	0.328
ML	1.329	1.521	3.19	0.71	2.4931	2.284	0.996	1.29175	-1.879	0.387	0.906	0.398
2Step	2.297	2.452	7.168	0.715	6.4661	4.783	0.999	3.78694	-1.781	0.41	0.82	0.415
TRUE	0.893	1.089	1.193	0.697	0.509	0.991	0.989	0.00537	-0.039	0.09	1.059	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.5, corr.var = -0.95, N = 800)												
SSLR	0.678	0.846	0.719	0.707	0.0261	1.486	0.996	0.49357	-0.819	0.194	1.21	0.332
ML	3.07	3.231	23.962	0.98	23.0001	21.941	1.273	20.67188	-1.968	Inf	1	Inf
2Step	5.483	5.588	71.921	0.748	71.1869	67.657	1.039	66.62108	-1.815	0.691	0.86	0.694
TRUE	0.906	1.096	1.206	0.699	0.5206	0.994	0.991	0.0058	-0.04	0.09	1.05	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.85, corr.var = 0, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.477	0.6	0.362	0.359	0.00917	2.416	1.044	1.37509	-1.624	0.145	1.9	0.216
ML	0.832	0.984	1.009	0.373	0.64251	1.203	0.996	0.21109	-2.047	0.206	1.05	0.236
2Step	0.795	0.947	0.928	0.372	0.56322	1.232	0.997	0.2382	-2.056	0.208	1.04	0.236
TRUE	1.107	1.254	1.577	0.412	1.17279	0.982	0.98	0.00549	-0.026	0.087	1.03	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.85, corr.var = -0.5, N = 800)												
SSLR	0.495	0.628	0.397	0.391	0.0131	2.223	0.981	1.24481	-1.383	0.145	1.514	0.235
ML	1.107	1.252	1.587	0.398	1.1953	0.99	0.968	0.02557	-1.854	0.223	0.876	0.254
2Step	1.753	1.867	3.744	0.411	3.3405	1.7	0.969	0.73424	-1.759	0.244	0.815	0.261
TRUE	1.111	1.256	1.584	0.402	1.1889	0.967	0.964	0.00538	-0.039	0.089	1.063	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.85, corr.var = -0.85, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.48	0.612	0.378	0.375	0.0099	2.268	0.983	1.28894	-1.308	0.144	1.317	0.245
ML	1.43	1.554	2.875	0.378	2.5041	1.569	0.983	0.58913	-2.166	Inf	1.182	Inf
2Step	2.78	2.849	9.033	0.382	8.6582	4.648	0.985	3.66615	-1.895	0.414	0.937	0.419
TRUE	1.16	1.29	1.67	0.375	1.3014	0.979	0.977	0.00486	-0.029	0.089	1.047	0.15
- generate_normal(seed = 22, censoring = 1.9, corr.errors = -0.85, corr.var = -0.95, N = 800)												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.487	0.623	0.391	0.388	0.0102	2.285	0.977	1.31E+00	-1.209	0.141	1.118	0.245
ML	3.932	4.03	37.989	0.639	37.3619	34.152	1.28	3.29E+01	-1.852	Inf	0.896	Inf
2Step	7.676	7.724	171.169	0.465	170.7128	161.682	1.075	1.61E+02	-1.636	0.681	0.696	0.685
TRUE	1.173	1.307	1.713	0.383	1.3363	0.979	0.977	5.73E-03	-0.031	0.089	1.043	0.152

FIGURE 9.7: Predictions Scenario 2, small sample size, high censoring degree ($\sim 80\%$)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring=0.35,corr.errors=0,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.803	1	1.01	1.01	0.00448	0.997	0.997	0.00356	-0.013	0.056	0.99	0.093
ML	0.803	1	1.01	1.01	0.00567	0.999	0.997	0.0047	-0.345	0.079	0.872	0.108
2Step	0.803	1	1.01	1.01	0.00572	0.999	0.997	0.00473	-0.344	0.079	0.871	0.108
TRUE	0.802	1	1.01	1.01	0.0042	0.997	0.997	0.0036	-0.034	0.045	1.033	0.07
censoring=0.35,corr.errors=0,corr.var=-0.5,N=2300												
SSLR	0.806	1	1.01	1.01	0.00448	0.984	0.984	0.00335	-0.036	0.056	1.013	0.093
ML	0.807	1	1.01	1.01	0.00628	0.986	0.984	0.00509	-0.312	0.107	0.855	0.122
2Step	0.807	1	1.01	1.01	0.00572	0.985	0.984	0.00456	-0.312	0.107	0.855	0.122
TRUE	0.807	1	1.01	1.01	0.00444	0.984	0.984	0.0032	0.01	0.045	0.949	0.07
censoring=0.35,corr.errors=0,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.807	1	1.01	1.01	0.00377	0.988	0.989	0.00296	-0.029	0.054	1.007	0.092
ML	0.811	1.01	1.02	1.01	0.0106	0.996	0.989	0.01038	-0.224	0.187	0.768	0.194
2Step	0.809	1.01	1.01	1.01	0.0073	0.993	0.989	0.00717	-0.225	0.187	0.768	0.194
TRUE	0.808	1	1.01	1.01	0.0036	0.988	0.988	0.0029	0.016	0.045	0.936	0.07
censoring=0.35,corr.errors=0,corr.var=-0.95,N=2300												
SSLR	0.805	1.01	1.01	1.01	0.0041	1	1	0.0037	-0.033	0.053	1.007	0.092
ML	0.813	1.02	1.03	1.01	0.02475	1.03	1	0.02608	-0.017	0.314	0.561	0.319
2Step	0.812	1.02	1.03	1.01	0.02479	1.03	1	0.02617	-0.009	0.319	0.552	0.323
TRUE	0.805	1.01	1.01	1.01	0.00445	1	1	0.00382	0.017	0.045	0.937	0.078
censoring=0.35,corr.errors=-0.5,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.761	0.95	0.905	0.904	0.00476	1.07	1.02	0.0607	-0.344	0.054	1.188	0.088
ML	0.774	0.974	0.95	0.906	0.04807	1.02	1.01	0.00478	-0.339	0.079	0.874	0.107
2Step	0.776	0.977	0.956	0.907	0.05354	1.02	1.01	0.00499	-0.332	0.079	0.859	0.107
TRUE	0.775	0.976	0.955	0.905	0.05403	1.01	1.01	0.00375	-0.03	0.045	1.03	0.07
censoring=0.35,corr.errors=-0.5,corr.var=-0.5,N=2300												
SSLR	0.76	0.947	0.899	0.898	0.00444	1.06	1	0.05636	-0.298	0.052	1.104	0.088
ML	0.782	0.979	0.961	0.899	0.06646	1	1	0.00438	-0.381	0.106	0.946	0.121
2Step	0.78	0.977	0.956	0.899	0.06173	1	1	0.00453	-0.38	0.108	0.946	0.123
TRUE	0.778	0.975	0.952	0.899	0.05758	1	1	0.0036	-0.005	0.046	0.981	0.07
censoring=0.35,corr.errors=-0.5,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.764	0.951	0.907	0.907	0.00366	1.06	1.01	0.05393	-0.233	0.051	0.983	0.087
ML	0.78	0.976	0.954	0.907	0.05154	1.01	1.01	0.00564	-0.162	0.181	0.721	0.189
2Step	0.774	0.969	0.94	0.907	0.0373	1.01	1.01	0.00863	-0.16	0.187	0.717	0.194
TRUE	0.78	0.977	0.956	0.907	0.05347	1	1.01	0.0029	0.002	0.046	0.969	0.078
censoring=0.35,corr.errors=-0.5,corr.var=-0.95,N=2300												
SSLR	0.764	0.955	0.914	0.913	0.00452	1.07	1.02	0.05221	-0.216	0.05	0.952	0.086
ML	0.781	0.98	0.962	0.913	0.05318	1.03	1.02	0.01383	-0.045	0.307	0.611	0.312
2Step	0.782	0.981	0.965	0.913	0.05599	1.05	1.02	0.02669	-0.03	0.319	0.595	0.323
TRUE	0.783	0.984	0.97	0.913	0.06066	1.02	1.02	0.0041	0.003	0.046	0.968	0.07
censoring=0.35,corr.errors=-0.85,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.677	0.843	0.712	0.711	0.00345	1.19	1.04	0.15529	-0.567	0.047	1.333	0.078
ML	0.731	0.93	0.868	0.718	0.1527	1.03	1.03	0.0041	-0.323	0.072	0.909	0.104
2Step	0.725	0.923	0.854	0.718	0.1388	1.03	1.03	0.00551	-0.31	0.079	0.861	0.107
TRUE	0.726	0.925	0.858	0.717	0.14416	1.03	1.03	0.0038	-0.024	0.046	1.026	0.07
censoring=0.35,corr.errors=-0.85,corr.var=-0.5,N=2300												
SSLR	0.666	0.832	0.694	0.693	0.004	1.17	1.03	0.15113	-0.501	0.046	1.208	0.077
ML	0.722	0.92	0.848	0.695	0.157	1.02	1.02	0.00416	-0.347	0.093	0.943	0.113
2Step	0.714	0.91	0.83	0.694	0.139	1.02	1.02	0.00511	-0.338	0.108	0.934	0.123
TRUE	0.719	0.916	0.841	0.695	0.15	1.02	1.02	0.004	-0.018	0.046	1.013	0.07
censoring=0.35,corr.errors=-0.85,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.668	0.833	0.696	0.696	0.00354	1.17	1.02	0.14722	-0.407	0.044	1.027	0.075
ML	0.72	0.92	0.849	0.696	0.15655	1.02	1.02	0.00357	-0.302	0.153	0.863	0.165
2Step	0.709	0.905	0.822	0.696	0.1295	1.03	1.02	0.00641	-0.265	0.187	0.822	0.194
TRUE	0.716	0.915	0.839	0.696	0.14624	1.02	1.02	0.0031	-0.012	0.046	1.002	0.079
censoring=0.35,corr.errors=-0.85,corr.var=-0.95,N=2300												
SSLR	0.663	0.829	0.689	0.689	0.00307	1.18	1.04	0.14665	-0.388	0.043	0.988	0.075
ML	0.721	0.922	0.853	0.689	0.16693	1.04	1.04	0.00473	-0.367	0.258	0.935	0.265
2Step	0.71	0.907	0.826	0.689	0.13956	1.05	1.04	0.01478	-0.258	0.318	0.822	0.322
TRUE	0.716	0.917	0.844	0.689	0.15789	1.04	1.04	0.0044	-0.01	0.046	0.999	0.07

FIGURE 9.8: Predictions Scenario 2, big sample size, low censoring degree ($\sim 25\%$)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring=1.9,corr.errors=0,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.815	1	1.01	1	0.0243	1.005	0.998	0.01104	-0.115	0.124	1.039	0.186
ML	0.889	1.12	1.26	1.01	0.2656	1.363	1.005	0.36192	-1.967	0.102	0.984	0.118
2Step	0.882	1.11	1.24	1.01	0.2451	1.336	1.004	0.33518	-1.969	0.101	0.985	0.118
TRUE	0.817	1	1.01	1	0.0247	0.997	0.997	0.0036	-0.034	0.045	1.033	0.077
censoring=1.9,corr.errors=0,corr.var=-0.5,N=2300												
SSLR	0.803	0.991	0.987	0.985	0.0191	0.988	0.984	0.00671	-0.064	0.109	0.978	0.18
ML	0.81	1.002	1.01	0.985	0.0421	1.026	0.984	0.04458	-1.86	0.127	0.816	0.133
2Step	0.809	1.001	1.008	0.985	0.0403	1.023	0.984	0.04155	-1.86	0.127	0.817	0.133
TRUE	0.802	0.988	0.981	0.98	0.0184	0.984	0.984	0.0032	0.01	0.045	0.949	0.078
censoring=1.9,corr.errors=0,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.806	0.991	0.991	0.991	0.0178	0.993	0.99	0.0061	-0.121	0.105	1.096	0.183
ML	0.827	1.019	1.051	0.991	0.0788	1.058	0.99	0.07119	-1.825	0.211	0.78	0.213
2Step	0.825	1.016	1.045	0.991	0.072	1.047	0.99	0.06004	-1.825	0.21	0.781	0.212
TRUE	0.808	0.992	0.991	0.99	0.0185	0.988	0.988	0.0029	0.016	0.045	0.936	0.078
censoring=1.9,corr.errors=0,corr.var=-0.95,N=2300												
SSLR	0.811	1	1.02	1.02	0.0196	1.01	1.01	0.0073	-0.149	0.103	1.133	0.183
ML	0.898	1.11	1.26	1.02	0.262	1.26	1.01	0.25754	-1.885	0.349	0.873	0.35
2Step	0.881	1.09	1.21	1.02	0.2092	1.18	1.01	0.17335	-1.893	0.349	0.88	0.35
TRUE	0.814	1.01	1.02	1.01	0.0225	1	1	0.0038	0.017	0.045	0.937	0.07
censoring=1.9,corr.errors=-0.5,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.73	0.906	0.824	0.822	0.0143	1.55	1.02	0.53129	-0.906	0.106	1.334	0.16
ML	0.8	0.982	0.97	0.826	0.1563	1.13	1.02	0.12086	-1.857	0.097	0.939	0.115
2Step	0.805	0.989	0.984	0.827	0.1699	1.12	1.02	0.10984	-1.856	0.097	0.937	0.115
TRUE	0.929	1.132	1.288	0.827	0.474	1.01	1.01	0.00375	-0.03	0.045	1.03	0.07
censoring=1.9,corr.errors=-0.5,corr.var=-0.5,N=2300												
SSLR	0.71	0.89	0.798	0.792	0.0182	1.54	1.01	0.54159	-0.848	0.094	1.195	0.156
ML	0.801	0.994	0.995	0.792	0.216	1.1	1	0.10084	-1.785	0.124	0.763	0.131
2Step	0.798	0.99	0.988	0.792	0.2088	1.11	1	0.10543	-1.783	0.125	0.76	0.131
TRUE	0.939	1.148	1.327	0.791	0.5487	1	1	0.0036	-0.005	0.046	0.981	0.078
censoring=1.9,corr.errors=-0.5,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.71	0.886	0.792	0.79	0.015	1.56	1.01	0.5563	-0.817	0.09	1.113	0.157
ML	0.908	1.109	1.249	0.79	0.472	1.05	1.01	0.04738	-1.887	0.206	0.896	0.208
2Step	0.903	1.105	1.242	0.79	0.465	1.05	1.01	0.04952	-1.882	0.207	0.89	0.209
TRUE	0.955	1.162	1.358	0.789	0.583	1	1.01	0.00299	0.002	0.046	0.969	0.078
censoring=1.9,corr.errors=-0.5,corr.var=-0.95,N=2300												
SSLR	0.718	0.895	0.806	0.803	0.017	1.55	1.02	0.53425	-0.788	0.089	1.079	0.158
ML	0.981	1.195	1.449	0.804	0.659	1.08	1.03	0.05964	-2.175	0.341	1.188	0.342
2Step	0.978	1.191	1.445	0.804	0.655	1.1	1.03	0.0732	-2.163	0.348	1.173	0.348
TRUE	0.952	1.166	1.367	0.799	0.582	1.02	1.02	0.00415	0.003	0.046	0.968	0.07
censoring=1.9,corr.errors=-0.85,corr.var=0,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.564	0.705	0.501	0.499	0.00961	2.4	1.05	1.35742	-1.407	0.081	1.464	0.124
ML	0.911	1.093	1.201	0.51	0.69892	1.12	1.03	0.09052	-1.874	0.097	0.885	0.115
2Step	0.957	1.139	1.306	0.512	0.80163	1.09	1.03	0.06314	-1.865	0.098	0.882	0.115
TRUE	1.145	1.32	1.748	0.51	1.24659	1.03	1.03	0.0038	-0.024	0.046	1.026	0.07
censoring=1.9,corr.errors=-0.85,corr.var=-0.5,N=2300												
SSLR	0.547	0.684	0.472	0.47	0.00874	2.4	1.03	1.37357	-1.31	0.072	1.252	0.118
ML	1.017	1.194	1.433	0.47	0.97029	1.06	1.02	0.03946	-1.741	0.118	0.792	0.128
2Step	0.951	1.13	1.284	0.47	0.8215	1.1	1.02	0.07786	-1.753	0.123	0.791	0.13
TRUE	1.172	1.34	1.803	0.471	1.34002	1.02	1.02	0.004	-0.018	0.046	1.013	0.079
censoring=1.9,corr.errors=-0.85,corr.var=-0.85,N=2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.527	0.663	0.444	0.443	0.00833	2.45	1.02	1.42945	-1.226	0.067	1.037	0.115
ML	1.047	1.221	1.499	0.443	1.06337	1.06	1.02	0.03776	-1.893	0.195	0.948	0.199
2Step	0.872	1.047	1.112	0.443	0.67713	1.19	1.02	0.16762	-1.955	0.207	0.996	0.208
TRUE	1.202	1.365	1.87	0.441	1.43571	1.02	1.02	0.0031	-0.012	0.046	1.002	0.079
censoring=1.9,corr.errors=-0.85,corr.var=-0.95,N=2300												
SSLR	0.528	0.664	0.444	0.443	0.00831	2.47	1.04	1.43567	-1.21	0.066	1.001	0.115
ML	1.108	1.279	1.644	0.444	1.20685	1.06	1.04	0.02084	-2.16	0.31	1.239	0.313
2Step	0.922	1.097	1.224	0.444	0.78686	1.18	1.04	0.14371	-2.23	0.344	1.304	0.345
TRUE	1.215	1.377	1.902	0.441	1.46801	1.04	1.04	0.0044	-0.01	0.046	0.999	0.07

FIGURE 9.9: Predictions Scenario 2, big sample size, high censoring degree (~80%)

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = -0.15, corr.errors = 0, SSLR N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.811	0.993	0.986	0.984	0.00697	1.02	1.01	0.00547	-0.064	0.108	1.084	0.176
ML	0.929	1.132	1.373	1.081	0.29741	1.44	1.11	0.33401	0.674	Inf	-0.027	Inf
2Step	1.508	1.76	3.791	1.255	2.54149	4.03	1.29	2.75073	0.674	0.12	-0.027	0.211
TRUE	0.81	0.991	0.984	0.983	0.00589	1.01	1.01	0.0051	-0.052	0.088	1.066	0.15
censoring = -0.15, corr.errors = -0.5, SSLR N = 800												
SSLR	0.758	0.926	0.858	0.856	0.00638	1.05	0.999	0.05394	-0.356	0.101	1.25	0.165
ML	0.896	1.093	1.293	0.973	0.32478	1.471	1.11	0.36472	0.171	Inf	1.01	Inf
2Step	1.32	1.545	2.923	1.045	1.88189	2.991	1.171	1.82424	0.166	0.115	1.02	0.214
TRUE	0.783	0.951	0.905	0.858	0.05033	0.998	0.996	0.0052	-0.04	0.087	1.05	0.15
censoring = -0.15, corr.errors = -0.85, SSLR N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.648	0.792	0.628	0.626	0.00477	1.15	0.998	0.15525	-0.643	0.087	1.5	0.142
ML	0.848	1.038	1.132	0.817	0.31849	1.237	1.108	0.13187	0.098	Inf	1.12	Inf
2Step	1.628	1.851	4.198	0.972	3.23008	3.691	1.213	2.48258	0.097	0.115	1.12	0.214
TRUE	0.723	0.882	0.779	0.643	0.13839	0.982	0.98	0.0054	-0.026	0.087	1.03	0.15
censoring = 1.4, corr.errors = 0, SSLR N = 800												
SSLR	0.742	0.92	0.85	0.843	0.0231	1.02	1.02	9.13E-03	-0.102	0.22	1.217	0.323
ML	8.635	9.257	184.346	23.129	161.6281	251.08	24.12	2.27E+02	-1.427	Inf	0.952	Inf
2Step	20.081	20.347	877.969	15.126	863.1122	986.71	15.73	9.71E+02	-1.412	0.143	0.952	0.229
TRUE	0.734	0.908	0.828	0.829	0.0149	1.01	1.01	5.11E+00	-0.052	0.088	1.066	0.15
censoring = 1.4, corr.errors = -0.5, SSLR N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
censoring = 1.4, corr.errors = -0.85, SSLR N = 800												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.463	0.587	0.347	0.343	0.0105	2.49	1.04	1.45E+00	-1.642	0.142	1.867	0.214
ML	6.749	7.19	109.234	12.793	96.6673	148.83	14.63	1.34E+02	-1.372	Inf	0.848	Inf
2Step	15.811	15.975	505.026	7.575	497.5858	569.899	9	5.61E+02	-1.361	0.142	0.855	0.229
TRUE	1.146	1.283	1.653	0.39	1.2697	0.982	0.98	5.49E+00	-0.026	0.087	1.034	0.15

FIGURE 9.10: Predictions Scenario 3, small sample size, low and high censoring degree

	Predictions in 'test_ws'					Predictions in 'test'			Parameters			
censoring = -0.15, corr.errors = 0, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.807	1.01	1.01	1.01	0.00454	0.998	0.997	0.00386	-0.027	0.057	0.996	0.093
ML	0.859	1.07	1.19	1.03	0.16043	1.182	1.016	0.16969	0.163	Inf	0.922	Inf
2Step	1.32	1.57	2.61	1.13	1.48689	2.689	1.123	1.56885	0.163	0.058	0.919	0.106
TRUE	0.806	1.01	1.01	1.01	0.00431	0.997	0.997	0.0036	-0.034	0.045	1.033	0.07
censoring = -0.15, corr.errors = -0.5, corr.var = 0, N = 2300												
SSLR	0.758	0.948	0.901	0.901	0.00439	1.08	1.02	0.06072	-0.353	0.053	1.206	0.088
ML	0.782	0.981	0.973	0.907	0.07107	1.05	1.02	0.03583	0.182	Inf	0.873	Inf
2Step	1.095	1.326	1.857	0.962	0.89991	1.68	1.06	0.62955	0.183	0.058	0.872	0.106
TRUE	0.776	0.975	0.953	0.903	0.05434	1.01	1.01	0.0037	-0.03	0.045	1.03	0.07
censoring = -0.15, corr.errors = -0.85, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals	Intercept	SE_Intercept	Var1	SE_Var1
SSLR	0.664	0.826	0.684	0.684	0.00322	1.2	1.05	0.16069	-0.604	0.047	1.393	0.076
ML	0.859	1.071	1.191	0.735	0.46003	1.21	1.05	0.15737	0.159	Inf	0.948	Inf
2Step	1.087	1.309	1.795	0.763	1.03559	1.56	1.07	0.49622	0.164	0.058	0.933	0.107
TRUE	0.716	0.915	0.839	0.693	0.14882	1.03	1.03	0.0038	-0.024	0.046	1.026	0.07
censoring = 1.4, corr.errors = 0, corr.var = 0, N = 2300												
SSLR	0.81	0.992	0.99	0.987	0.02	1.001	0.998	6.70E-03	-0.054	0.131	0.974	0.194
ML	3.495	3.847	46.11	6.404	39.8144	63.778	7.888	5.59E+01	-1.427	Inf	1.05	Inf
2Step	23.354	23.659	586.291	15.049	571.4989	683.65	19.303	6.64E+02	-1.431	0.074	1.065	0.118
TRUE	0.808	0.989	0.985	0.983	0.0189	0.997	0.997	3.63E-03	-0.034	0.045	1.033	0.077
censoring = 1.4, corr.errors = -0.5, corr.var = 0, N = 2300												
Model	MAE	RMSE	MSE	Var.residuals	Sqr_bias.residuals	MSE	Var.residuals	Sqr_bias.residuals				
SSLR	0.725	0.898	0.809	0.808	1.44E-02	1.62	1.02	5.96E-01				
ML	10.968	11.894	190.207	28.384	1.62E+02	245.12	33.85	2.11E+02				
2Step	32.202	32.606	1087.445	27.04	1.06E+03	1200.44	32	1.17E+03				
TRUE	0.939	1.151	1.332	0.811	5.34E-01	1.01	1.01	3.75E-03				
censoring = 1.4, corr.errors = -0.85, corr.var = 0, N = 2300												
SSLR	0.537	0.673	0.456	0.454	9.74E-03	2.56	1.06	1.51E+00	-1.534	0.08	1.593	0.121
ML	4.673	5.02	39.163	4.866	3.44E+01	40.44	5.73	3.47E+01	-1.304	Inf	0.923	Inf
2Step	14.51	14.683	231.288	5.382	2.26E+02	225.51	5.99	2.20E+02	-1.308	0.07	0.942	0.114
TRUE	1.175	1.341	1.806	0.473	1.34E+00	1.03	1.03	3.86E-03	-0.024	0.046	1.026	0.079

FIGURE 9.11: Predictions Scenario 3, big sample size, low and high censoring degree.